

Stable Finite SID

Theory, adversarial synthetic evidence, failure diagnosis, and decision rules

Internal methods synthesis for the diffusionCircuit project

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Abstract

The project has accumulated several methods that initially look separate: quadratic score-based interaction discovery, the original score-based temporal graph operator, point-history tangents, finite history contrasts, conditional diffusion, and channel-specific changepoint scans. They are better understood as components of one program. The population object is the observed conditional predictive law $h \mapsto P(Y \mid H = h)$. A predictive backend represents that law by a normalized density, an outcome score, a joint-state score, or samples. An adapter converts the representation into a supported history contrast or derivative. A typed readout declares whether the target is mean, dispersion, covariance, interaction, skewness, or tail behavior. A calibrated inference layer can then detect static structure or temporal changes.

This synthesis audits the recent synthetic experiments under that decomposition and then replaces the weakest synthetic lane with exact, adversarial conditional laws. The empirical conclusion is candid. Normalized likelihood and classical sparse dynamics remain the leaders when their models are tractable and approximately correct. Outcome-space denoising score matching, history-score SID, and flow velocity are different mathematical objects; a good denoising loss does not by itself estimate SID. The earlier neuromodulator stress suite also contains decisive implementation and comparison defects, including a non-operative E3 shift, deterministic “stochastic” release, marginal rather than joint density scoring, and unequal method cell sets. Its rankings are therefore diagnostic rather than adjudicative.

The recommended research direction is *finite, typed, support-aware SID*: form an exact finite signed contrast at a scientifically declared perturbation scale; estimate bounded or projected witnesses through multiple routes; and report channel-specific functionals with route-agreement and mechanism-confusion controls. On the new exact whole-future-path law, a raw path classifier recovers a pure fourth-order effect to 3.05% median error while linear and quadratic routes recover none. Full-law diffusion, flow, and autoregressive samplers recover only 0–6% of that effect. Classifier guidance of a frozen midpoint generator is a partial repair—EDM rises from 1% to 40% effect slope—but does not replace the higher-order geometry the generator discarded. Explicit endpoint experts under the same total data budget also help EDM, but only to 34% slope. The evidence therefore favors a normalized global mixture gate or finite ratio/classifier head *and* topology-aligned expert training, with diffusion or flow used only for residual within-mode geometry. Exact-rejection bounded energy, frozen-anchor stochastic interpolation, and an 80-cell endpoint-expert factorial make that conclusion testable rather than architectural speculation.

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1 July 13 adversarial revision: what changed

Bottom line. The main problem was not one thing. The quadratic SID failures are partly structural; the first neuromodulator stress suite also had invalid or missing experimental axes; and several comparisons conflated predictive-law backends with SID estimators. The repaired program keeps finite typed SID as the estimand, uses normalized densities, bounded energies, diffusion, and flow only as alternative backends, and tests all routes on exact laws designed to conceal changes from mean and covariance baselines.

Four corrections govern the rest of this report.

1. **Derivative types are not interchangeable.** Diffusion DSM targets $\nabla_y \log p_\sigma(y | h)$; infinitesimal SID targets $D_v \log p(y | h)$; flow matching targets a transport velocity in artificial time. A diffusion or flow can support finite SID through samples, but its native vector field is not itself the SID witness.
2. **The original quadratic DSM parameterization is inadmissible.** If an affine noisy score has precision B_σ , a positive clean Gaussian covariance requires $0 \prec B_\sigma \prec \sigma^{-2}I$. Free affine natural parameters do not enforce this spectral interval and cannot remain in a bounded interval globally over an unbounded history space.
3. **The earlier E3/E4 neuromodulator evidence is downgraded.** The E3 reservoir branch ignored the declared state-mixture shift, so its two OOD levels were bit-identical. E4's release was deterministic conditional on state. Neural and classical rankings also averaged unequal cell sets, and response RMSE was scaled by overall signal variation rather than the intervention-effect scale.
4. **A real home field must hide changes from low-order baselines.** The new exact path law changes only higher-order multimodal topology along its declared contrast direction; its conditional mean and covariance are invariant. Gaussian methods therefore cannot win merely because the target is mean-like, while exact density, score, ratio, and sampling oracles remain available.

1.1 The revised method stack

The operative scientific object is

predictive-law backend $\longrightarrow$ finite signed contrast $\longrightarrow$ typed readout $\longrightarrow$ dynamic/interventional aggregation.

The layers have different validation obligations. A held-out NLL or energy score validates a backend; witness calibration validates a contrast adapter; channel RMSE validates a typed readout; and rollout or changepoint error validates aggregation. No native training loss substitutes for the downstream gate.

1.2 Bounded finite SID theorem

For histories $h_1, \dots, h_R$, signed coefficients c_r with $\sum_r c_r = 0$, and $C = \sum_r |c_r|$, define

$$\nu_c = \sum_r c_r P_{h_r}, \quad M_c = \sum_r \frac{|c_r|}{C} P_{h_r}, \quad W_c = \frac{d\nu_c}{dM_c}.$$

Then $\nu_c \ll M_c$ even if the contrasted laws are mutually singular, $|W_c| \leq C$, $E_{M_c} W_c = 0$, and

$$\sum_r c_r E_{P_{h_r}} \phi(Y) = E_{M_c} \{\phi(Y) W_c(Y)\}$$

for every integrable ϕ . If a label R is drawn with $\Pr(R = r) = |c_r|/C$ followed by $Y \sim P_{h_r}$, then the calibrated posterior $\pi_r(y) = \Pr(R = r | Y = y)$ gives

$$W_c(y) = C \sum_r \text{sign}(c_r) \pi_r(y).$$

For a central first difference this becomes

$$W_\delta(y) = \delta^{-1} \{2\eta(y) - 1\} = \delta^{-1} \tanh\left\{\frac{1}{2} \log(p_+(y)/p_-(y))\right\}.$$

Thus the recommended finite witness is a bounded transform of a potentially unbounded density ratio. With excess classifier log loss $\mathcal{R}_{\log}$, Pinsker's inequality yields

$$E_{M_c}(\widehat{W}_c - W_c)^2 \leq 2C^2 \mathcal{R}_{\log},$$

which makes calibration a quantitative SID diagnostic rather than an optional classifier statistic. The ratio-class-probability link is also the basis of proper-composite density-ratio estimation [13].

1.3 Direct readouts and common-random-number coupling

For one predeclared ϕ , the direct group-mean contrast is generally more efficient than learning an omnibus witness. The omnibus witness earns its cost when many readouts are queried, response-space localization is itself scientific, or one shared backend interpolates many histories. Generators also permit a useful coupling: with $Y_\pm = G(h_\pm, Z)$ and the same latent Z ,

$$\widehat{\Delta}_{\text{pair}} = \frac{1}{n} \sum_i \frac{\phi\{G(h_+, Z_i)\} - \phi\{G(h_-, Z_i)\}}{2\delta}.$$

If the paired numerator is $O_{L^2}(\delta)$, the variance is $O(n^{-1})$ rather than the generic independent-sample $O\{(n\delta^2)^{-1}\}$. This coupling, not the native diffusion loss, is one plausible advantage of flow and diffusion backends.

1.4 When classifier guidance can repair a generator

For a balanced two-history contrast, let $M = (P_+ + P_-)/2$ and $\eta(y) = \Pr(Z = + | Y = y)$. The exact identities

$$\frac{dP_+}{dM} = 2\eta, \quad \frac{dP_-}{dM} = 2(1 - \eta)$$

give a constructive global-mass head. If a generator draws exactly from M , self-normalized reweighting by $\widehat{\eta}$ and $1 - \widehat{\eta}$ recovers the two endpoint laws when the classifier is calibrated. This is the finite-SID analogue of guidance, but it has a sharper target than multiplying a conditional score difference.

The construction also separates two errors. Write Q for the learned midpoint law, with densities q, m , and let $\widehat{P}_+ \propto \widehat{\eta}q$. If $A = \int \widehat{\eta}q \geq \kappa > 0$, then

$$\text{TV}(\widehat{P}_+, P_+) \leq \kappa^{-1} \|\widehat{\eta}q - \eta m\|_1 \leq \kappa^{-1} \left\{ E_Q |\widehat{\eta} - \eta| + \int \eta |q - m| \right\}.$$

The first term is contrast-head error; the second is midpoint-generator geometry error. Hence even a Bayes classifier cannot recover a fourth-order component that is absent from Q . Conversely, fitting a perfect generator without a contrast-aligned head leaves the global mass derivative unidentified in the separated-support limit. This decomposition motivates a statistical mixture-of-transport rather than post-hoc guidance alone.

1.5 A stable score-to-SID bridge

For an energy law $p_\theta(y | h) \propto \exp\{-E_\theta(y, h)\}$,

$$D_v \log p_\theta(y | h) = -D_v E_\theta(y, h) + E_{p_\theta}\{D_v E_\theta(Y, h)\}.$$

The partition function cancels after conditional centering. The implemented tail-safe version starts from a normalized Gaussian reference q_ψ and uses

$$p_{\theta, \psi}(y | h) \propto q_\psi(y | h) \exp\{B \tanh f_\theta(y, h)\}.$$

It inherits the reference tails, has importance weights in $[e^{-B}, e^B]$, is curl-free in response space, and exposes a centered history derivative. This is a direct theoretical bridge from non-normalized score modeling to SID. An unconstrained denoiser does not have the same property.

The distinction between the law and its numerical sampler is essential. The initial implementation draws a finite pool from q_ψ and resamples using softmax weights. That is sampling–importance resampling: it is finite-pool biased and repeated outputs share the proposal pool. Its draws therefore do not satisfy the independence assumption used by the fair finite-ensemble energy score, and its current proper-score rows are excluded from primary rankings. A bounded tilt admits an exact replacement. If $Y^\star \sim q_\psi(\cdot | h)$ and $U \sim \text{Unif}(0, 1)$, accept when

$$\log U \leq B \tanh f_\theta(Y^\star, h) - B.$$

Accepted proposals are i.i.d. from $p_{\theta, \psi}$, with acceptance $e^{-B} Z(h) \geq e^{-2B}$. Thus B controls a genuine approximation–computation tradeoff: larger B reduces ratio clipping but can make exact sampling and importance centering exponentially inefficient. Confirmation requires exact rejection draws, conditional acceptance and proposal-cap diagnostics, and agreement between exact centering and a separately reported self-normalized importance estimate.

The exact sampler is now implemented and validated against analytic tilted Gaussians. Reevaluation of all 64 frozen bounded-energy checkpoints completes without failure: empirical median acceptance is 0.136–0.431 across laws and proposal materialization never exceeds the 4,096 cap. Its fair energy is competitive but not generally best, and its G8 quartic slope is essentially zero. Thus the bridge is mathematically and computationally viable, but the current ratio-versus-Gaussian training objective still does not allocate signal to the hard conditional mass channel.

An objective-controlled score comparison also freezes one Gaussian reference and fits the same bounded tilt by conditional Hyvärinen loss,

$$E\left[\frac{1}{2}\|s_q + \nabla_y u\|^2 + \nabla_y \cdot s_q + \Delta_y u\right], \quad u = B \tanh f_\theta.$$

This isolates clean response-score fitting from classifier ratio fitting. It is expected to favor smooth within-mode shape changes, whereas a calibrated ratio classifier should retain an advantage when only separated component masses change. Neither objective is licensed as a history-SID estimator until the centered history derivative is validated downstream.

1.6 Why SBTG can be intrinsically ill-conditioned

SBTG’s ideal mixed object $K_v = D_v \nabla_y \log p_h = \nabla_y \ell_v$ recovers only the response gradient of the history witness. If P_h satisfies a Poincaré inequality with constant C_P and a centered reconstruction obeys $\|\widehat{\nabla \ell} - K_v\|_{L^2(P_h)} \leq \epsilon$, then

$$\|\widehat{\ell} - \ell_v\|_{L^2(P_h)} \leq \sqrt{C_P} \epsilon.$$

Separated mixtures can have enormous C_P ; disconnected support leaves one unknown constant per component. Consequently, clean outcome scores may barely change when only mixture weights change. This is a theoretical home-field for finite classifiers and ratios, and a structural limitation of score-gradient reconstruction rather than merely a training failure.

The limiting case makes the obstruction exact. Let

$$p_a(y) = \sum_{k=1}^K \pi_k(a) f_k(y), \quad \text{supp}(f_j) \cap \text{supp}(f_k) = \emptyset \quad (j \neq k),$$

where the component laws are fixed and only their history-dependent masses $\pi_k(a)$ vary. For $y \in \text{supp}(f_k)$,

$$\nabla_y \log p_a(y) = \nabla_y \log f_k(y), \quad \partial_a \log p_a(y) = \partial_a \log \pi_k(a).$$

Thus every history has exactly the same clean outcome-score field even though its SID witness is nonzero and can be large. Clean Hyvärinen or denoising score matching cannot infer the missing component constants from within-component gradients. With Gaussian corruption, the supports overlap and the obstruction becomes an ill-conditioning result rather than non-identification: the conditional score difference is concentrated in the low-probability frame-ambiguity region. Increasing σ first exposes that region and then erases the original law contrast. A generic average over noise levels can therefore achieve a low DSM loss while allocating almost no effective signal to history-controlled mode mass.

This proposition also explains why the appropriate repair is not simply a larger denoiser. A full backend for this regime needs a global-mass channel: a normalized mixture gate, a calibrated finite classifier, a density-ratio or bounded-energy head, or an equivalent contrast-aligned auxiliary loss. Diffusion or flow can still model the conditional geometry within components, but the gate or ratio head must carry the component constants that local outcome scores discard.

1.7 Stabilization implemented in the new benchmark

The new backend panel includes: (i) clean-covariance Gaussian parameterization trained by NLL or multiscale DSM; (ii) an exact conditional affine flow; (iii) EDM input/output/skip preconditioning, log-normal noise sampling, a $\rho = 7$ schedule, and Heun integration; (iv) EDM anchored around a learned conditional Gaussian mean and scale; (v) ordinary and Gaussian-source conditional flow matching; and (vi) the bounded energy tilt above. These changes follow the separation and preconditioning principles of EDM [14], the simulation-free vector regression of flow matching [15], and the Gaussian-reference geometry needed for function-space diffusion [16]. Whole-path operator methods are reserved for an actual trajectory-valued response, not used merely because the conditioner is a time series.

The literature suggests several additional stabilizers, but they address different failure modes. Stable Target Field replaces a single noisy DSM target by an importance-weighted reference-batch average and specifically reduces target variance in the ambiguous intermediate-noise regime [17]. On G8 it should be tested with reference batches stratified by declared control; it can reduce variance near the visibility peak, but it cannot recover component masses from clean within-component gradients. Data-dependent or Gaussian-process source laws shorten the transport and preserve temporal covariance, as in TSFlow [18]; a single Gaussian source still matches only first- and second-order structure and therefore needs a ratio or mixture-gate head for the pure fourth-order lane. Denoising diffusion operators supply the correct Gaussian-reference/Cameron–Martin geometry for whole functions [16], but do not change the statistical estimand. Stochastic interpolants supply a non-singular conditional transport and post-training diffusion choice [19], but likewise require a contrast diagnostic to establish that conditioning is actually used.

These observations define a minimal stabilization factorial rather than an unstructured architecture search:

1. EDM preconditioning and ordinary loss weighting;
2. topology-window noise stratification, centered near the empirically visible intermediate scales;
3. the same schedule with a control-stratified Stable Target Field;
4. an auxiliary balanced finite-contrast classifier or bounded energy-ratio head, cross-fitted and calibrated;
5. a statistical mixture-of-transport backend with a normalized history-dependent gate and diffusion/flow experts within modes.

Items 2–3 test numerical/statistical stabilization of the local score. Items 4–5 add the global mass information that the separated-support proposition says is absent from that score. Only the latter can be expected to solve a pure mode-weight SID problem in the disjoint-support limit. A classifier-free guidance scale by itself is not a substitute: multiplying a nearly zero conditional score difference does not create the missing per-component constants.

The controlled transport experiment uses a conditional point-source stochastic interpolant, following the forecasting construction of Chen et al. [19]. With a frozen Gaussian conditional-mean anchor $a(h)$, standardized response y , and $\xi \sim N(0, I)$, compare only

$$I_t = (1 - t)a(h) + \beta_t y + (1 - t)\sqrt{t}\xi, \quad R_t = -a(h) + \dot{\beta}_t y - \sqrt{t}\xi,$$

for $\beta_t = t$ versus $\beta_t = t^2$. The latter has $\dot{\beta}_0 = 0$ and is designed to remove a difficult initial transport impulse. Its native sampler is the Euler–Maruyama SDE

$$dX_t = b_\theta(X_t, t, h) dt + (1 - t)dW_t, \quad X_0 = a(h).$$

The apparently unusual $-\sqrt{t}\xi$ target is the forward-SDE drift, not the raw time derivative of the interpolation noise. With $\gamma_t = (1 - t)\sqrt{t}$ and diffusion $g_t = 1 - t$, the score correction gives $\dot{\gamma}_t \xi - g_t^2 \xi / (2\gamma_t) = -\sqrt{t}\xi$. The implementation and tests lock this identity, $\epsilon = 1$, and the 128-step primary protocol. This is a sampler-backend experiment, not a new SID definition. The primary comparison uses 128 steps with 64/256-step resolution checks. All 32 frozen-anchor

fits completed. Squared β improves G8 fair energy from 0.717 to 0.710 but reduces the already tiny quartic effect slope from 0.0075 to 0.0007; all three sampler resolutions agree. On the Gaussian G1 lane both schedules retain a real mean effect (NRMSE 0.338/0.319). Thus the G8 result is not an unstable sampler or initial transport impulse. Better transport conditioning improves a coarse law metric while leaving the missing conditional mass statistic unresolved.

A final endpoint-expert factorial tests whether an explicit normalized gate alone is sufficient. It crosses four systems, two data seeds, two model seeds, and five backend families. Each method receives 8,000 total training examples, split 4,000/4,000 between the supported $-1/+1$ experts; the declared control is removed from expert inputs and a deterministic gate selects the endpoint expert. All 80 cells complete. Relative to shared balanced-control fits, slopes increase to 0.336 for EDM, 0.251 for affine flow, 0.179 for MDN, and 0.035 for flow matching; Gaussian remains zero. This establishes that explicit gating helps, but the best NRMSE is still 0.665. The next architecture therefore needs both a normalized gate and a contrast/topology-aligned expert loss; ordinary proper-law or transport losses remain insufficient at this budget.

1.8 Exact whole-path falsification law

The new G8 response is a flattened multichannel future path. Let $U \in \mathbb{R}^{d \times r}$ be orthonormal traveling-wave motifs and let

$$\mathcal{Z}_A = \{\pm\sqrt{r}e_i\}_{i=1}^r, \quad \mathcal{Z}_B = \{\pm\sqrt{r}Re_i\}_{i=1}^r$$

for a non-axis-aligned orthogonal R . Both frames have mean zero and covariance I_r . Given nonlinear observed history h , a stable path mean $m(h)$, correlated noise C_0 , and a declared shape control $a(h)$, G8 uses

$$Y \mid h, z \sim N\{m(h) + \alpha U z, C_0\}, \quad \Pr(z \in \mathcal{Z}_B \mid h) = \text{logit}^{-1}\{1.2a(h)\}.$$

Changing the declared control changes mixture topology and higher moments while mean and covariance remain exactly invariant. The law is a finite Gaussian mixture with exact clean/noisy log density, response score, history tangent, finite ratio, and conditional sampler. This is the decisive synthetic lane for whether a flexible joint-law backend and finite SID adapter can expose scientifically typed path changes that Gaussian baselines erase.

The readout must match the construction. A dictionary ending at cubic order is blind by design to much of this symmetric, moment-matched signal. The corrected panel includes an exact frame-occupancy readout and a noise-adjusted fourth-order motif contrast. With $u = U^\top\{Y - m(h)\}$, use a contrast of terms

$$H_4(x; s^2) = x^4 - 6s^2x^2 + 3s^4$$

in the original and rotated motif frames. Linear and quadratic path features are predeclared negative controls, not weak positive targets. An exact Gaussian shadow $N\{m(h), C_0 + \alpha^2 U U^\top\}$ is run beside G8 with the same histories; it matches G8's conditional first two moments but has identically zero shape contrast. In eight balanced-classifier negative-control cells its median AUC is 0.499 (range 0.496–0.505), and the largest spurious classifier-projected quartic effect is 0.16% of the corresponding positive G8 effect. The successful G8 classifier is therefore not reading a mean/covariance mismatch or the control label itself.

The first nonlinear-measurement draft is not admissible evidence: it corrupts only future outputs while giving every model the latent history, and it restarts the calcium, bleaching, and artifact processes at the prediction origin. The repaired experiment generates joint measured histories and futures, carries measurement state across the boundary, appends missingness and time-since-observation, and separates a declared-operator transfer task from an unannounced target-law shift. Because $p(Y \mid H_{\text{obs}})$ is not analytically available, this lane uses proper scores on independent joint episodes and population randomized-operation contrasts; pointwise oracle SID remains confined to the clean latent lane.

1.9 What backend error can and cannot guarantee

For a reverse diffusion with scalar diffusion coefficient g_t , common terminal law, exact score s_t , and estimated score $\hat{s}_t$, the usual Girsanov conditions give a path-space KL proportional to

$$\frac{1}{2} \int g_t^2 E \|s_t - \hat{s}_t\|^2 dt.$$

Data processing and Pinsker therefore bound terminal expectation error for bounded $|\phi| \leq B$ by

$$\left| E_{P_0} \phi - E_{\hat{P}_0} \phi \right| \leq B \left\{ \int g_t^2 E \|s_t - \hat{s}_t\|^2 dt \right\}^{1/2}.$$

This supports integrated, noise-weighted score diagnostics for sampling. It does not imply accurate $D_h \log p$, and a small native DSM average can conceal the intermediate-noise scales that carry mixture-weight information.

For flow matching, couple the true and learned ODEs with the same source draw. If both velocities are L -Lipschitz, Grönwall’s inequality yields a Wasserstein endpoint bound of the form

$$W_2(P_h, \hat{P}_h) \leq \int_0^1 e^{L(1-t)} \{E \|v_t(X_t, h) - \hat{v}_t(X_t, h)\|^2\}^{1/2} dt.$$

Lipschitz readouts inherit this error. Again, velocity MSE alone does not control history derivatives of the flow. The primary SID interface remains paired finite samples unless normalized CNF likelihood is actually integrated.

1.10 Function-space scope

The finite signed-measure definition does not require Euclidean response space; M_c and $W_c = d\nu_c/dM_c$ exist on abstract path and function spaces. A function-space diffusion is more delicate: there is no infinite-dimensional Lebesgue density, the score is defined relative to a Gaussian reference in Cameron–Martin geometry, and the corruption covariance must be trace class. Denoising diffusion operators therefore motivate GP/AR corruption and neural operators only when Y is an entire future trajectory or field. They do not make an ordinary one-step vector SID estimator more identified. This distinction is why the new benchmark first uses a finite exact path grid, then adds a sample-only nonlinear measurement view and reserves resolution-transfer tests for a later operator experiment.

1.11 Frozen advancement rules

A flexible generative backend advances only if it improves a held-out proper law score over the strongest normalized baseline, beats both the autoregressive density and conditional flow on a hard law, remains non-inferior on the Gaussian sanity law, and has stable sampler resolution. A score-derived SID adapter advances only if it normalizes, centers, maintains usable importance-sampling ESS, satisfies the SID moment identity, and improves on direct paired channels or calibrated classifier ratios for at least two non-mean readouts. Otherwise the generator may remain useful while the score-based SID claim is rejected.

2 Executive verdict

Bottom line. There is no evidence that an outcome-score or diffusion objective is currently the best general estimator of conditional dynamics. The best-developed original idea is finite, typed, support-aware SID. It now succeeds not only for location and low-rank occupancy changes but also for the audited whole-path fourth-order G8 effect when trained as a calibrated finite classifier. The same effect is almost absent from full-law diffusion, flow, and autoregressive samplers despite excellent marginal likelihood. The immediate research program is therefore a contrast-aligned ratio/energy or normalized mixture-gate head, with diffusion or flow used for within-mode path geometry when that flexibility is actually needed.

The main conclusions are:

1. **The methods are not six independent theories.** They share one population object, $P_h = P(Y \mid H = h)$, and differ mainly in representation, adapter, readout, and inference layer.
2. **SID should be the umbrella, not the quadratic implementation.** The current closed-form quadratic model is denoted SID-Q. It is one convenient score-matching backend with Gaussian reconstruction, not the definition of SID.

3. **Finite contrasts are the most promising core.** They define an exact, supported comparison of predictive laws and can be estimated from densities, density ratios, classification, scores, or samples. This is the natural generalization of infinitesimal SID.
4. **SBTG is a derivative/localization route, not a separate mechanism estimator.** Its ideal mixed operator is $K_v = D_v \nabla_y \log p_h = \nabla_y D_v \log p_h$. Recovering the history witness from this gradient requires integration, centering, support, and boundary conditions. Current empirical SBTG statistics are reduced-form screens and are not mechanism-specific.
5. **Diffusion and stochastic interpolation are backends.** They are useful if the law is complex, the normalizer is unavailable, or sampling is required. Denoising score matching targets a corrupted outcome score and stochastic interpolation a transport drift; neither by itself identifies the clean history derivative or a physical mechanism. The new interpolant is stable on G1 yet has essentially zero G8 quartic slope at every tested sampler resolution.
6. **Changepoints are an inference wrapper.** A changepoint scan can be applied to a validated mean, variance, tail, or general-law statistic. It does not rescue an invalid static channel.
7. **Current wins are narrow but real.** Finite signed classifiers/ratios recover location changes, ratio critics recover low-rank mixture interactions, and a contrast-aligned classifier recovers the G8 fourth-order path effect within 1–3% median error. Classifier guidance partially repairs all five frozen G8 generators, but the best guided slope is only 0.40; explicit endpoint experts reach only 0.34. A mass head and a geometry-preserving, contrast-aligned generator are both needed. Mean and matched-tail changepoint probes work strongly in matched settings.
8. **Current failures are structural, not just optimization failures.** Covariance and skew adapters fail even with oracle samples; local outcome scores are ill-conditioned for separated component masses; score risk does not control history-derivative error; generic gating readouts lose to the zero field; and SBTG dispersion statistics barely exceed chance.
9. **A full generator is optional, not mandatory.** For one declared contrast/readout, a direct classifier or projected moment can be substantially more accurate and efficient. A shared generator is justified only by proper-law quality, many-query amortization, sample-only access, or a downstream need for realistic trajectories.

2.1 What should carry the project forward

The project should be organized around the following stack:

$$\begin{aligned} \text{predictive law/generator} &\longrightarrow \text{finite supported contrast} \longrightarrow \text{bounded or projected witness} \\ &\longrightarrow \text{typed channel readout} \longrightarrow \text{calibrated inference.} \end{aligned}$$

The specific working hypothesis is not “diffusion beats likelihood.” It is:

When the conditional law is sufficiently non-Gaussian or only available through a sampler/score, finite supported contrast adapters can expose scientifically useful history sensitivities that simple mean models and generic two-sample tests cannot localize, while retaining controlled claims and calibration.

That hypothesis remains plausible, but it has not yet been demonstrated in the hard channels or in a regime where the flexible backend is necessary.

3 One mathematical object, four layers

3.1 The population object

Let $H \in \mathcal{H}$ denote an observed history or conditioning state and $Y \in \mathcal{Y} \subseteq \mathbb{R}^d$ the future response. The basic object is the conditional predictive law

$$P_h(A) = \Pr(Y \in A \mid H = h), \quad A \subseteq \mathcal{Y}.$$

When a density exists, write $p_h(y)$. This object is observed-law and predictive. Without further intervention, invariance, sufficiency, or structural assumptions, it is not automatically a causal mechanism.

Every current non-latent method can be placed in four layers.

Layer	Question	Examples in the project
Representation	How is P_h learned or accessed?	Gaussian/Student- t /MDN density; conditional outcome score; joint-state score; diffusion sampler; oracle simulator
Adapter	How are two nearby histories compared?	Autodifferentiation; finite density ratio; signed classifier; Riesz projection; score difference; mixed SBTG operator
Readout	Which aspect of the response law is the target?	Mean, variance, covariance, low-rank interaction, skewness, tail probability, local occupancy
Inference	How is uncertainty or temporal change declared?	Bootstrap, held-out route agreement, CUSUM/GLR, calibrated windows, change-point localization

This separation prevents three recurrent category errors: comparing a graph score with a conditional density by one scalar; treating a native training loss as proof of downstream derivative accuracy; and interpreting a detected distributional change as a specific mechanism.

3.2 Infinitesimal history sensitivity

Let $T_{\epsilon,v}h$ perturb history h in a declared direction v . If differentiability and common-support conditions hold, define the history score or infinitesimal SID witness

$$\ell_v(y; h) = D_v \log p_h(y) = \left. \frac{\partial}{\partial \epsilon} \log p_{T_{\epsilon,v}h}(y) \right|_{\epsilon=0}.$$

It is conditionally centered, $\mathbb{E}_h[\ell_v(Y; h)] = 0$. For any integrable response feature ϕ ,

$$D_v \mathbb{E}_h[\phi(Y)] = \mathbb{E}_h[\phi(Y) \ell_v(Y; h)]. \quad (1)$$

Equation (1) is the main unifying identity. The witness ℓ_v describes how the entire conditional law changes. Choosing ϕ turns it into a typed scientific readout.

Examples include

$\phi_j(y) = y_j$	mean susceptibility,
$\phi_{jk}(y; h) = (y_j - \mu_j(h))(y_k - \mu_k(h))$	covariance susceptibility,
$\phi_{j,\tau}^{\text{tail}}(y; h) = \mathbf{1}\{ y_j - \mu_j(h) > \tau \sigma_j(h)\}$	shape/tail susceptibility,
$\phi_a^{\text{int}}(y) = a(y)$	declared interaction or occupancy channel.

The response dictionary is not cosmetic. A method may estimate a useful mean channel while completely missing covariance or skew. This is exactly what the synthetic experiments show.

3.3 Finite signed contrasts: the preferred estimand

Infinitesimal derivatives are fragile when the learned law is rough, the perturbation leaves support, or score convergence does not imply derivative convergence. For histories $h_1, \dots, h_R$ and signed coefficients c_r satisfying $\sum_r c_r = 0$, define

$$\nu_c = \sum_{r=1}^R c_r P_{h_r}.$$

Let $C = \sum_r |c_r|$ and choose the absolute-coefficient mixture

$$M_c = \frac{1}{C} \sum_{r=1}^R |c_r| P_{h_r}.$$

Then $\nu_c \ll M_c$, and the finite SID witness

$$W_c(y) = \frac{d\nu_c}{dM_c}(y)$$

is bounded by $|W_c(y)| \leq C$. For every integrable ϕ ,

$$\sum_{r=1}^R c_r \mathbb{E}_{h_r}[\phi(Y)] = \mathbb{E}_{M_c}[\phi(Y) W_c(Y)]. \quad (2)$$

This identity is exact at the declared resolution; it needs no Taylor approximation.

For a central first contrast, $h_{\pm} = T_{\pm\delta, v} h$ and $c = (1/(2\delta), -1/(2\delta))$. Higher-order parity stencils can reduce truncation bias, provided the corresponding signed measure and reference mixture are used consistently. The finite target is preferable for the near-term program because it is:

- scientifically interpretable at a declared perturbation magnitude δ ;
- support-aware and testable with oracle samples;
- accessible from normalized densities, density ratios, classification, scores, or paired generators;
- bounded under the absolute-mixture construction;
- compatible with the same typed readouts as the infinitesimal witness.

The infinitesimal witness should be treated as a limit or secondary analysis, not as the only definition of the program.

3.4 How SBTG fits

Define the conditional outcome score

$$s_y(y; h) = \nabla_y \log p_h(y).$$

The ideal mixed SBTG operator in history direction v is

$$K_v(y; h) = D_v s_y(y; h) = \nabla_y \ell_v(y; h). \quad (3)$$

Thus SBTG is not unrelated to SID: it observes the response-space gradient of the SID witness. But K_v does not determine ℓ_v without a reconstruction problem. One must specify support, boundary conditions, conditional centering, and a stable integration or Hodge/Poisson procedure. A local cross-moment of a learned joint score is therefore not automatically an estimator of ℓ_v , a physical field, or a causal graph.

The correct hierarchy is

$$P_h \longrightarrow \ell_v \longrightarrow K_v = \nabla_y \ell_v \longrightarrow \text{a declared localization or graph summary.}$$

Each arrow loses information or adds assumptions. The current experiments mainly validate the final reduced-form localization summary, and only weakly outside mean-like effects.

3.5 How diffusion fits

Conditional diffusion or denoising score matching learns a noisy response score such as

$$s_{\theta}(y_{\sigma}, h, \sigma) \approx \nabla_{y_{\sigma}} \log p_{\sigma}(y_{\sigma} \mid h),$$

where p_{σ} is the Gaussian-corrupted law. This representation can support reverse-time sampling and can model complex laws without a tractable normalizer [2, 3]. It does not directly equal ℓ_v , K_v , or a clean-law density ratio.

There are three legitimate diffusion-to-SID routes:

1. generate conditional samples at h_+ and h_- , then estimate a finite signed witness or channel contrast;
2. compare conditional noisy scores across histories and solve a validated reconstruction problem;
3. estimate channel expectations directly from paired conditional samples, possibly with common random numbers.

The first and third are the most robust. Calling diffusion a backend makes the framework future-proof: MDNs, flows, autoregressive samplers, energy models, and simulators can expose the same interface.

3.6 How changepoints fit

Let $Z_t^{(\phi)}$ be a validated channel statistic derived from the predictive residual or contrast. A changepoint method scans for a change in its distribution or mean. This produces a statement of the form “the declared tail channel changed near time τ , at the calibrated resolution.” It does not alter the population target or raise the causal claim ceiling.

The correct order is:

$$\begin{aligned} \text{static oracle validity} &\rightarrow \text{static learned-model validity} \rightarrow \text{matched-null calibration} \\ &\rightarrow \text{power/localization} \rightarrow \text{mechanism-confusion testing.} \end{aligned}$$

4 Consolidating the named methods

The table below replaces the earlier “six methods” view. The latent mechanistic state-space model is intentionally excluded from this synthesis, as requested; it is a different structural program with different identification assumptions.

Name	Representation	Target/adaptor	What it can claim	Current status
SID-Q	Quadratic conditional outcome score with Gaussian reconstruction	Hyvärinen or denoising score matching; analytic mean/log-variance readouts	Conditional and fitted Gaussian mean/dispersion sensitivities	Viable, but loses to Gaussian NLL and sparse dynamics on its home tasks
SBTG-K	Joint or conditional response score	Mixed derivative $D_v s_y$, followed by localization or reconstruction	Reduced-form sensitivity localization; full SID only after validated integration	Moderate mean-like screening; weak and non-specific dispersion/gating evidence
Point-history SID	Normalized density or score	Infinitesimal $D_v \log p_h$ or autodifferentiated history derivative	Local change of the predictive law under smoothness and support assumptions	Optimization converges, but covariance/skew readiness gates fail
FINITE-SID	Densities, ratios, classifiers, scores, or samples	Exact signed law contrast $W_c = d\nu_c/dM_c$	Supported finite law change and typed channel contrasts	Best original direction; useful on location and low-rank mixture tasks
DIFFUSION-SID	Conditional noisy score and sampler	Sample contrast, score difference, or pathwise paired readout	Same finite/typed targets if adapter is validated	Flexible backend; no current empirical advantage over likelihood
CP-SID	Any validated static channel statistic	Calibrated sequential/windowed scan	Detection/localization of a declared channel change	Mean and tail work; variance probe is not channel-specific

Proposed naming is deliberately modular:

- **SID** is the broad predictive-law sensitivity framework.
- **SID-Q** is the current strict quadratic, Gaussian-reconstructed implementation.
- **SBTG-K** is the mixed response-score-gradient/localization route.
- **FINITE-SID** is the recommended finite signed-contrast adapter.
- **DIFFUSION-SID** uses a diffusion/score sampler as the predictive backend.
- **CP-SID** applies calibrated change inference to a validated SID channel.

This naming makes the empirical comparison fair: the same backend can be paired with several adapters, and the same adapter can be paired with several backends.

5 What each objective actually identifies

5.1 Normalized likelihood

Negative log likelihood is a strictly proper objective for a normalized conditional law: in a sufficiently rich and correctly optimized class, it identifies P_h itself. It therefore exposes likelihood ratios, sampling, moments, and history derivatives by direct computation. Its weaknesses are model misspecification, normalization cost, poor tail parameterization, and derivative instability when the fitted law is not smooth in h .

When a low-dimensional Gaussian, Student- t , or mixture law is adequate, this is the home field of likelihood. The present synthetic results confirm that score matching should not be expected to dominate there.

5.2 Hyvärinen score matching

Score matching fits $\nabla_y \log p_h(y)$ without requiring the normalizing constant [1]. Under regularity and identification conditions it can recover an unnormalized model’s score. It is useful for energy-based models, but it controls response-score risk, not history-score risk. In particular,

$$s_{\theta,n} \rightarrow s_y \quad \not\Rightarrow \quad D_v s_{\theta,n} \rightarrow D_v s_y,$$

without stronger smoothness and derivative-uniform convergence. This gap is central to the weak gating and history-tangent results.

5.3 Denoising score matching

Denoising avoids second derivatives and estimates the score of a corrupted law. The corruption scale σ creates a real bias–variance tradeoff: larger σ regularizes but smooths fine distributional changes. The history perturbation δ and corruption σ are different resolutions and must be swept jointly. A good denoising loss does not prove recovery of a clean-law derivative.

5.4 Joint-state SBTG score

A joint score of standardized consecutive states targets $\nabla_{(x_t, x_{t+1})} \log p(x_t, x_{t+1})$. This differs from the conditional outcome score by marginal-history terms in the history component. Outcome-score components can coincide under careful partitioning, but the commonly used cross-moments and local graph summaries introduce additional choices. Their proper comparison is with reduced-form graph/localization baselines, not with conditional-law NLL.

5.5 Ratio and classification objectives

Direct density-ratio estimators such as KLIEP and uLSIF estimate p_+/p_- without separately fitting both densities [4]. Balanced classifiers estimate posterior class probabilities, which can be converted into ratios or bounded signed witnesses. These routes directly target finite law differences and often avoid differentiating a flexible model with respect to history.

Their weaknesses are overlap, class-separation saturation, calibration, and feature geometry. The covariance/skew failures with oracle samples show that architecture and response features can dominate predictive-model quality.

5.6 Moment/Riesz projection

If only selected channels matter, one can project the signed witness onto a dictionary $\psi(Y)$, or estimate a Riesz representer for the desired linear functional. Cross-fitting and debiasing can reduce regularization bias in downstream functionals [10]. This route may be more sample-efficient than reconstructing the entire witness, but it is only as general as the dictionary and can be ill-conditioned. The current signed Riesz dictionary failed the hard channels and requires redesign, not merely retuning.

6 Alternative approaches and required baselines by subproblem

No single comparator set is appropriate for every layer. The following table defines the serious alternatives that should be present in future experiments.

Subproblem	Main alternatives	Mandatory baselines	SID opportunity and tradeoff
Conditional law	Gaussian/Student- t likelihood; mixtures; normalizing flows; autoregressive densities; energy models; diffusion	Ridge Gaussian, heteroskedastic Gaussian NLL, Student- t , MDN	Score/sample backends help when normalization or density evaluation is hard; otherwise likelihood is more direct and currently better
Conditional mean dynamics	VAR/ridge; sparse transition; SINDy; Gaussian process; neural autoregression	Persistence/zero, ridge VAR, Granger, SINDy	Typed SID can add distributional channels, but should not replace simpler mean estimators on mean-only tasks
Outcome-score estimation	Hyvärinen, sliced score matching, DSM, diffusion	Analytic oracle score; fitted Gaussian score; neural NLL score	Avoids normalizers and supports sampling; adds corruption choice and does not guarantee history derivatives
Finite law comparison	Direct log-density ratio; KLIEP/uLSIF; logistic classification; MMD; energy distance; optimal transport	Oracle density ratio, oracle-sample classifier, MMD and energy	Signed SID gives a localized witness and typed effects, whereas generic two-sample tests primarily answer whether laws differ
History derivative	Autodiff of normalized law; local polynomial derivatives; Stein identities; Riesz/DML; infinitesimal perturbation analysis	Oracle finite difference and oracle tangent; zero derivative	Finite SID avoids derivative instability but depends on perturbation direction, scale, overlap, and adapter geometry
Distributional channels	Direct moment regression; quantile regression; expectiles; distribution regression; tail regression	Direct empirical moment/quantile difference; zero field	One witness can share information across channels, but a universal witness may be harder than a targeted channel estimator

Subproblem	Main alternatives	Mandatory baselines	SID opportunity and tradeoff
Reduced-form graph	Lagged correlation; ridge/Granger; PCMCI; VAR-LiNGAM; NOTEARS/DYNOTEARS; transfer entropy	Ridge/Granger, sparse transition, PCMCI, VAR-LiNGAM	SBTG may capture nonlinear distributional dependence, but its graph requires assumptions and currently lacks mechanism specificity
Offline change-points	CUSUM/GLR/Chow; PELT; binary segmentation; kernel MMD; energy distance	Mean CUSUM, residual variance/tail probes, MMD, energy, PELT	Typed SID may identify what changed; calibration and nuisance-confusion are harder than generic detection
Online changes	Sequential likelihood ratios; BOCPD; e-values/martingales; conformal monitoring	BOCPD and calibrated CUSUM	A bounded finite witness is attractive for sequential concentration, but dependence and repeated testing must be handled
Long-horizon response	Recursive autoregression; direct multi-horizon prediction; neural ODE/SDE; pathwise couplings; conditional bridges	Direct multi-horizon ridge/MDN; empirical paired rollout	Pathwise SID may decompose response across horizons, but compounding model error and coupling choice are substantial

Some baseline families address distinct questions:

- MMD is an omnibus two-sample discrepancy over an RKHS function class [5]; it can detect a change without identifying a response-space witness with the desired scientific interpretation.
- SINDy targets sparse governing equations in a chosen feature library [6]; it is a strong mean-dynamics baseline, not a full conditional-law estimator.
- PCMCI, VAR-LiNGAM, and DYNOTEARS add different causal/discovery assumptions [7, 8, 9]. They should be reported as observational structure learners under their own assumptions, not as interchangeable causal truth machines.
- PELT and Bayesian online changepoint detection solve segmentation or run-length inference [11, 12]; they can consume SID-derived costs but do not define the channel being monitored.

7 Empirical evidence: what works and what does not

7.1 Evidence standard and comparability

Legacy-evidence warning. The objective-parity, finite-contrast, graph, and changepoint results below remain useful because their recorded estimands and common-cell comparisons are explicit. The later E1–E5 neuromodulator “rethink” rankings are not used to adjudicate methods after the July 13 code audit: E3’s advertised state-mixture shift was non-operative, E4’s release was not stochastic, density scores were primarily marginal/factorized, and method averages used unequal selected cells. Section 13 is the replacement evidence table.

The central audit rule is: compare methods on a common estimand, common data-generating process, common seed intersection, common information set, and common metric. Native loss is secondary. The main field metrics are normalized integrated squared error, where the zero-field baseline is 1, and witness NRMSE, where the zero-witness baseline is also 1. Values above 1 are worse than predicting no effect.

The newest matched objective panel contains 250 of 250 completed cases: ten methods, five mechanisms, and fresh DGP seeds 8–12. The finite-contrast rerun contains 76,760 valid metric rows across ten adapter/evaluation seeds. The latter tests adapter randomness around one rebuilt generator/model source; it does not constitute ten independent generator fits.

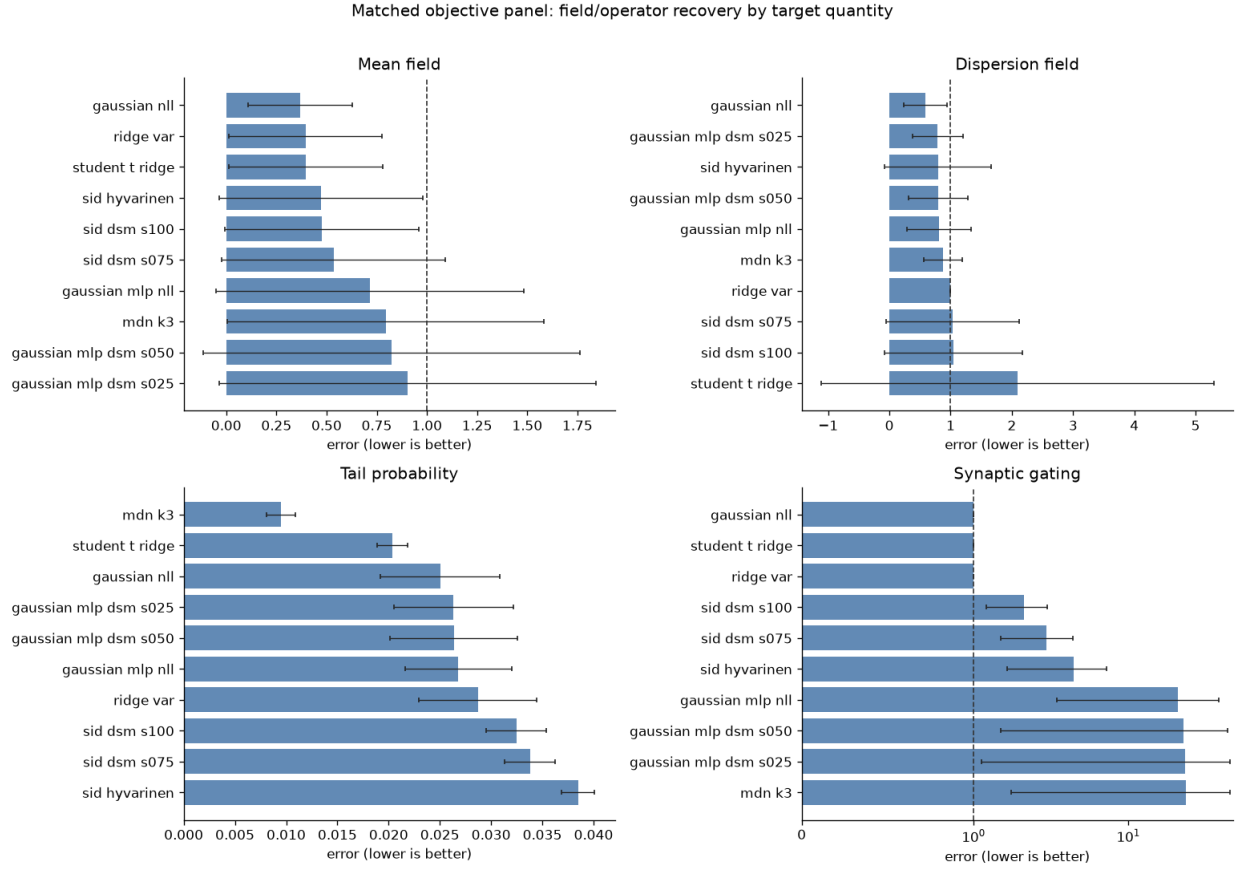


Figure 1: Matched objective comparison. Methods should be read by estimand: held-out law quality, mean-field recovery, and dispersion-field recovery are different tasks.

7.2 Predictive law and physical fields

For the physical additive-mean field in the new matched panel:

Method	Mean-field nISE
Penalized Gaussian NLL	0.369
Ridge Gaussian	0.395
Student- <i>t</i> ridge	0.395
Quadratic Hyvärinen SID	0.473
Quadratic DSM SID	0.475–0.535
Neural Gaussian NLL	0.716
MDN	0.794
Neural Gaussian DSM	0.824–0.902

An earlier mechanism panel was even more favorable to classical dynamics: SINDy scored 0.194, sparse transition 0.201, Gaussian NLL 0.246, and Hyvärinen SID 0.268. The exact values differ across panels, but the ordering is consistent.

For the physical log-variance field:

Method	Dispersion-field nISE
Penalized Gaussian NLL	0.586
Neural Gaussian DSM, $\sigma = .25$	0.789
Quadratic Hyvärinen SID	0.791
Neural Gaussian DSM, $\sigma = .50$	0.794
Neural Gaussian NLL	0.809
MDN	0.872
Zero/constant-variance ridge	1.000
Quadratic DSM SID	1.037–1.048
Student- t ridge	2.091

Neural DSM modestly improves over neural NLL within the same broad Gaussian architecture, so denoising may be a useful derivative regularizer. It does not beat the strictly parameterized Gaussian likelihood. This is evidence for a regularization effect, not a general score-objective advantage.

On the matched-tail law, representational flexibility matters:

Method	Held-out NLL	Tail-probability RMSE
MDN, $K = 3$	-0.614	0.0095
Student- t ridge	-0.530	0.0204
Gaussian NLL	-0.288	0.0250
Neural Gaussian DSM	-0.280 to -0.283	0.0263–0.0264
Quadratic DSM SID	about -0.283	0.0325–0.0338
Quadratic Hyvärinen SID	9.18	0.0385

Yet the MDN’s physical shape-tail derivative has $\text{nISE} = 2.02$, worse than the zero derivative. Predicting a tail probability and estimating its derivative with respect to a modulator are distinct problems.

7.3 Synaptic gating: no current winner

All nonzero learned readouts are worse than the zero field:

Method	Total gating-field nISE
Gaussian NLL / ridge / Student- t , emit zero	1.000
Quadratic DSM SID	2.10–2.95
Quadratic Hyvärinen SID	4.42
Neural NLL / DSM / MDN	20.97–23.40

The trajectories may contain gating, but the present fitted-law readout does not identify its mixed derivative reliably. More hyperparameter search over the same target is unlikely to change the conclusion. The next experiment must first show oracle identification from paired controlled laws or an explicitly gated transition model.

7.4 History tangents and finite contrasts

The infinitesimal point-history sweep completed 96 of 96 fits, passed fixed-batch overfit checks, and reproduced under exact replay. Nevertheless, six derivative-readiness gates failed. G1 location tangents reached NRMSE 0.589–0.719, while the best G2 covariance and G3 skew tangents were 1.068 and 1.167. The failure is therefore not explained by gross optimization failure.

At clean $\delta = 0.1$, the ten-seed finite-contrast panel gives:

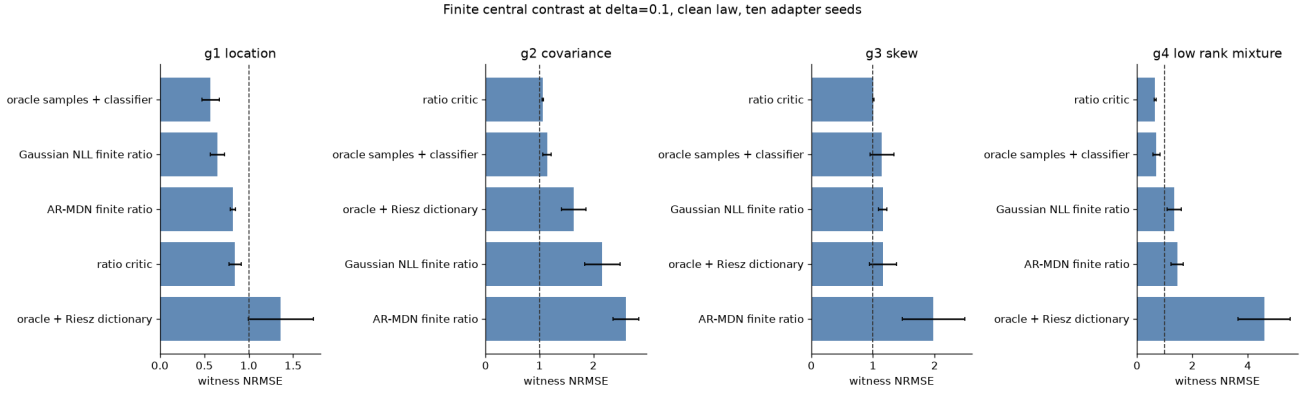


Figure 2: Finite-contrast witness NRMSE across ten adapter/evaluation seeds at the main perturbation scale. The dashed reference at 1 is the zero-witness baseline.

Generator	Best practically relevant route	Mean NRMSE	Verdict
G1 location	Direct signed classifier / Gaussian finite ratio	about 0.61–0.65	Useful; beats zero clearly
G2 covariance-only	Ratio critic	1.057	No recovery
G3 skew-only	Ratio critic	1.008	No recovery
G4 low-rank mixture	Ratio critic	0.656	Useful; beats zero clearly

The oracle infinitesimal tangent has negligible Taylor mismatch at this δ for G2 and G3, but the oracle-sample classifier is still roughly 1.14 NRMSE. Consequently, the primary bottleneck is adapter geometry, response features, calibration, or signal-to-noise—not the learned diffusion generator. The current signed Riesz dictionary should not be scaled until an oracle redesign beats zero.

7.5 SBTG localization

The frozen SBTG panel shows joint-score cross-moment average-precision excess of roughly 0.15–0.39 for mean-like support. Similar values occur across additive, dispersion, and synaptic mechanisms, so the statistic is not mechanism-specific. Log-variance localization is only 0.06–0.10, and squared-score-covariance localization is roughly 0.00–0.05. Linear SBTG is often as good as or better than feature-bilinear variants.

The defensible conclusion is that SBTG can remain a reduced-form nonlinear screening/localization route. It has not established recovery of physical dispersion, gating, or a causal graph.

7.6 Classical graph baselines and hidden confounding

Under causal sufficiency, classical transition methods reach about 0.80 AP excess. Under a strong hidden common driver, VAR-LiNGAM scores 0.407, ridge VAR 0.362, conditional Granger 0.361, PCMCI 0.338, sparse transition 0.253, and marginal lagged correlation 0.106. These are mandatory comparators. Even the best result is observational support recovery under a violated or stressed assumption, not identification of a latent causal mechanism.

7.7 Changepoints

With 199 matched-null calibration trajectories and 100 independent evaluation trajectories per condition:

- the mean CUSUM detects and localizes 100/100 high-signal mean changes;
- the residual tail-shape probe detects and localizes 100/100 exactly mean/variance-matched tail changes;
- the residual variance probe calls 100/100 dispersion changes but localizes only 41/100;

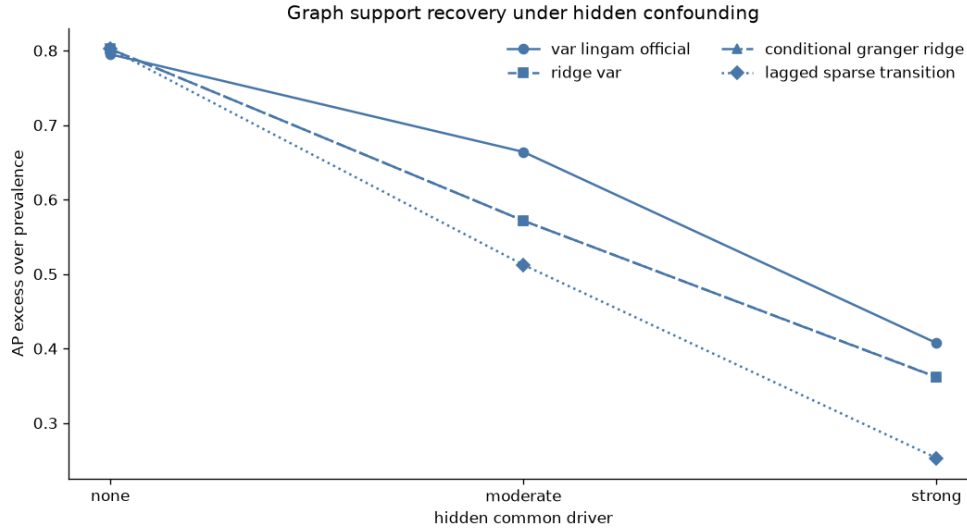


Figure 3: Graph-support average-precision excess under causal sufficiency and strong hidden common drive. Every observational method degrades.

- that variance probe also calls 95% of mean changes and 99% of matched-tail changes, so it is a broad residual-instability alarm rather than a dispersion-specific detector;
- independent-null false-positive rates are 2% for mean, 4% for conditional rule and variance, 3% for tail, 9% for MMD, and 11% for energy distance.

Mean and tail provide strong matched demonstrations. Dispersion requires nuisance orthogonalization and confusion controls before it can carry a channel-specific label.

8 Where our approaches have an advantage—and where they do not

8.1 Potential advantages

The unified finite SID approach can be better than existing alternatives for reasons that are conceptual and testable:

1. **Model-agnostic downstream interface.** The same contrast and readout can be applied to a tractable density, an energy model, a diffusion sampler, or an oracle simulator.
2. **Exact finite estimand.** Equation (2) avoids pretending that an arbitrarily small derivative is scientifically meaningful or statistically stable.
3. **Typed explanation of distributional change.** Unlike a single MMD or energy statistic, the witness can be integrated against declared mean, covariance, interaction, and tail features.
4. **Bounded witness construction.** The absolute-coefficient mixture gives a bounded Radon–Nikodym witness, which is attractive for stable estimation and sequential concentration.
5. **Multiple overidentifying routes.** Density ratios, classifiers, direct sample moments, and score reconstructions can target the same contrast. Agreement becomes a diagnostic rather than a rhetorical robustness check.
6. **Resolution as a scientific parameter.** δ specifies the perturbation being studied; σ specifies score corruption. Their effects can be separated and reported.
7. **A path from local sensitivities to changes over time.** Once a channel is validated, the same typed statistic can be embedded in offline or online changepoint inference.

8.2 Costs and failure modes

Those advantages come with real tradeoffs:

- **Perturbation design:** the direction, magnitude, stencil, and support of $T_{\delta,v}h$ are part of the estimand and may not be naturally identified from observational histories.
- **Overlap:** finite ratios and classifiers become unstable when P_{h+} and P_{h-} have little common support; tiny δ has the opposite weak-signal problem.
- **Channel conditioning:** covariance, skew, and tail features can have high variance or poor geometry. An omnibus witness may be harder to learn than a direct targeted regression.
- **Derivative instability:** score or density convergence does not automatically imply convergence of derivatives with respect to history.
- **Corruption bias:** DSM learns a smoothed law. Fine changes can be attenuated by the denoising scale.
- **No causality for free:** predictive history dependence can arise from common drive, selection, or omitted state.
- **Computation:** multiple histories, routes, channels, and calibration blocks cost more than a single fitted transition model.
- **Baseline strength:** when the task really is conditional mean estimation under a sparse low-order law, ridge or SINDy is expected to win.

8.3 When diffusion/score methods have a credible hope

The present negative results do not imply that diffusion-based SID is hopeless. They imply that it needs a home-field experiment. A score/sample backend has a credible advantage when at least one of the following holds:

- the conditional law is high-dimensional and multimodal, making normalized likelihood evaluation or integration impractical;
- the model is available only as a simulator or sampler;
- the scientific target depends on a local portion of response space that a mean/variance model erases;
- a shared generative model amortizes many contrasts or channels more efficiently than separately fitting targeted regressions;
- sample coupling or common noise can sharply reduce finite-difference variance.

To be impactful, the project must demonstrate one of these regimes. Beating a Gaussian likelihood on a near-Gaussian low-dimensional simulator is neither necessary nor likely. Conversely, merely functioning where likelihood is inconvenient is not enough; the method must yield a better estimand, better sample efficiency, or a capability the baseline cannot expose.

9 Theory agenda

The framework is conceptually coherent, but several results are needed to turn it into a mature method.

9.1 Finite-witness stability bounds

Derive error transfer from predictive laws or ratios to the witness and typed channels. Under overlap bounds and bounded ϕ , seek inequalities of the form

$$\left| \widehat{\Delta}_c(\phi) - \Delta_c(\phi) \right| \leq C_1 d(\widehat{P}, P) + C_2 \|\widehat{W} - W\|_{L^2(M)} + \text{sampling error},$$

with explicit dependence on δ , dimension, and overlap. This would clarify when improving the predictive backend can actually improve the scientific readout.

9.2 Bias–variance theory for δ and σ

Develop a phase diagram separating:

- finite-stencil truncation bias as δ grows;
- weak contrast and ratio variance as $\delta \rightarrow 0$;
- posterior smoothing/contraction as denoising noise σ grows;
- approximation and discretization error in reverse sampling.

The key theoretical question is whether the useful region scales with σ/δ , signal geometry, and intrinsic response dimension.

9.3 Orthogonal and cross-fitted channel estimation

Treat $\Delta_c(\phi)$ as the primary low-dimensional functional and derive Neyman-orthogonal estimators using cross-fitted predictive laws and Riesz representers. This may allow slower nuisance convergence while retaining valid uncertainty for a small set of scientific channels. It is likely more promising for covariance and tail readouts than reconstructing the entire witness with a generic classifier.

9.4 Reconstructing SID from SBTG

Formalize the inverse problem in Equation (3). Given an estimate of $K_v = \nabla_y \ell_v$, recover ℓ_v by a weighted Poisson or Hodge problem with the conditional centering constraint

$$\int \ell_v(y; h) p_h(y) dy = 0.$$

Prove identifiability and stability under boundary/support assumptions. Compare this reconstruction against direct finite ratios at the oracle level. If the inverse problem is ill-conditioned in the tested regimes, SBTG should remain a localizer rather than the main SID route.

9.5 Higher-order parity stencils

Develop first- and second-order signed stencils whose finite witnesses are derived from the correct likelihood-ratio polynomial. A second derivative of a density includes both log-density Hessian and score-product terms; using a log Hessian alone generally targets the wrong signed measure. Establish exact coefficient/reference-mixture constructions and variance-optimal weighting.

9.6 Route-agreement tests

When density, classifier, direct-moment, and score routes target the same $\Delta_c(\phi)$, formulate an overidentification test. Systematic disagreement can diagnose lack of overlap, score non-integrability, density calibration error, or sampler bias. This is potentially a distinctive contribution: the framework creates several estimators of one declared object rather than several incomparable model scores.

9.7 Mechanism-confusion and nuisance invariance

For a channel label to be meaningful, derive or enforce orthogonality to nuisance changes. A dispersion statistic should be insensitive to mean shifts; a standardized tail statistic should be insensitive to matched mean and variance changes. Develop residualization, influence-function, or adversarial nuisance-projection schemes and evaluate a full confusion matrix rather than only target power.

9.8 Sequential inference

Use bounded finite witnesses to construct blockwise e-values, martingales, or self-normalized scans under temporal dependence. The target result is false-alarm control across repeated monitoring while retaining a typed interpretation. Offline PELT and online BOCPD remain external inference engines; SID should contribute a validated cost or evidence sequence.

10 The next decisive experiment program

The next work should be staged. Later phases are conditional on earlier oracle gates; otherwise capacity and compute will obscure identification failures.

10.1 Observable neuromodulator benchmark ladder

The attached neuromodulatory hierarchy is useful if it is translated into observable conditional laws rather than another latent mechanistic state-space model. Let H_t contain observed stimulus, neural, behavioral, and neuromodulator histories, and let $Y_{t+1:t+K}$ be a multichannel future path. Five nested generator tiers should share histories and random-number streams:

Tier	Declared change	Construction and exact negative controls
O1	mean/gain	nonlinear saturating temporal operator with fixed covariance; variance/tail readouts are null
O2	stochastic law	heteroskedastic, Student- t , stochastic-release, and mixture residuals matched in mean; location is null
O3	source–target–timescale tensor	sparse multiscale history filters with known low-rank interaction tensor; matched marginal histories and indirect-path controls
O4	spatial field	smooth and localized kernels over uncertain coordinates, with matched global gain and spatially permuted nulls
O5	whole-path topology	G8-type mixture-of-path transports whose occupancy, dwell, or transition law changes while first two conditional moments remain fixed

The nonlinear backbone, residual law, and contrast mechanism must be independently randomized. A method should not receive the basis or architecture used to generate the law. Recommended primary sizes are 32/96/256 observed channels, history lengths 16/64, future horizons 8/32, and intrinsic contrast ranks 2/8/16, with at least 20 independent system seeds for any confirmatory claim. Every tier exposes exact conditional samples, paired latent randomness, and an exact density or high-precision oracle ratio where tractable.

Process noise and measurement noise are separate axes. Process axes include release randomness, heavy-tailed innovations, history-dependent scale, shared drive, and temporal/spatial correlation. Observation axes include independent noise at 0/0.1/0.25/0.5/1 signal SD, calcium-kernel misspecification at 0/10/25/50%, downsampling by 1/2/4/8, random and structured missing channels at 0/25/50/75%, shared artifacts, spatial-coordinate error, and intervention-time jitter. Single-axis sweeps establish attribution; a frozen composite stress condition tests robustness. Training-support interpolation and extrapolation are reported separately.

The seven scientific endpoints are predictive-law/calibration quality, perturbation fidelity, gain, variability/covariance, source–target influence stability, temporal/spatial kernel recovery, and path-transition/dwell effects. For each endpoint the direct sufficient-statistic regression or empirical paired contrast is mandatory; a calibrated finite classifier, kernel/MMD or energy two-sample method, appropriate normalized density, and diffusion/flow backend receive the same sample and tuning budget. This design tests whether a shared generative backend earns its complexity under misspecification, rather than only whether it can rediscover its own simulator.

10.2 Phase 0: oracle adapter tournament

Goal. Determine whether each scientific channel is recoverable from perfect access to the relevant conditional laws before training any flexible predictive model.

Data-generating families.

- G1: pure location change;
- G2: covariance-only rotation/scale change with fixed mean;
- G3: skew-only change with matched mean and covariance;
- G4: low-rank mixture interaction;
- G8: whole-path topology with exact first/second-moment shadow;
- mechanistic fields: additive mean, stochastic dispersion, matched tail, synaptic gating, and correlation-only changes.

Oracle interfaces. Exact log density and ratio where available; exact score; independent conditional samples; paired/common-random-number samples; exact moments; exact infinitesimal tangent.

Adapters.

- exact finite ratio and direct log-ratio transformation;
- calibrated signed logistic classifier;
- KLIEP and uLSIF;
- polynomial/Hermite, kernel, and learned feature classifiers;
- channel-targeted Riesz estimators with cross-fitting;
- score-difference plus weighted Poisson/Hodge reconstruction;
- mixed SBTG operator plus the same reconstruction;
- direct paired-sample channel difference.

Design. At least 20 independent generator seeds, with separate data, adapter, and evaluation seeds. Sweep sample size, dimension, δ , overlap, and signal strength. The main perturbation scale is fixed before inspecting hard-channel results; other scales are sensitivity analyses.

Primary metrics. Witness NRMSE and channel nISE against the zero baseline of 1, with confidence intervals over generator seeds. Secondary metrics are calibration, centering, route disagreement, runtime, and effective sample size.

Gate. A channel advances only if at least one practical oracle adapter has an upper 95% confidence bound below 1 and the intended channel is distinguishable from nuisance alternatives. G2/G3 currently fail this gate.

10.3 Phase 1: backend $\times$ adapter $\times$ channel factorial

Goal. Separate predictive-model error from adapter error and determine whether a shared flexible backend helps across multiple channels.

Axis	Levels
Backend	Oracle; ridge Gaussian; heteroskedastic Gaussian NLL; Student- t ; MDN; SID-Q-Hyvärinen; SID-Q-DSM; conditional score/diffusion; joint-score SBTG
Adapter	Direct density ratio; signed classifier; direct sample channels; Riesz; score/Hodge; mixed SBTG/Hodge
Channel	Mean; variance; covariance; low-rank interaction; skew; tail; local occupancy
Regime	Correct specification; heavy tails; mixtures; response coupling; high dimension; sample-only access

Every cell is evaluated on the same common estimand. Native NLL or DSM loss is reported only as a backend diagnostic. Exact seed intersections and identical train/validation/evaluation histories are required.

Success criterion for diffusion. A diffusion/score backend must either (i) beat the best feasible normalized backend on a common channel with uncertainty, or (ii) solve a sample-only/intractable-density task where the normalized comparator genuinely cannot expose the estimand, while beating direct empirical and generic two-sample baselines.

10.4 Phase 2: scaling and stress map

Goal. Find the regime, if any, where the unified approach has a real advantage.

Sweep:

- sample size and effective dependent-block count;
- response dimension, history length, and intrinsic contrast rank;
- perturbation δ and denoising scale σ ;
- observation noise, multimodality, tail weight, and support overlap;
- model misspecification and hidden common drive;
- number of requested channels/contrasts, to test amortization.

Report phase diagrams, not only aggregate ranks. The important question is where each method crosses NRMSE 1, how quickly it improves with data, and whether sharing a backend across many contrasts changes total cost.

10.5 Phase 3: mechanism-confusion battery

For every claimed channel, construct matched nuisances:

Claimed channel	Positive condition	Mandatory negative/confusion conditions
Mean	mean shift	variance, skew, tail, correlation changes
Dispersion	variance/scale change	mean, matched-tail, mixture-weight changes
Tail/shape	tail change matched in mean/variance	mean, variance, outlier contamination
Interaction	low-rank coupling change	marginal mean/variance changes
Graph/localization	edge-specific conditional change	common drive, indirect path, autocorrelation

Report a confusion matrix of call rates and localization, not only target AUROC or AP.

10.6 Phase 4: changepoints only for validated channels

Carry forward only Phase 0–3 winners. Compare:

- channel-specific CUSUM/GLR and PELT costs;
- MMD and energy-distance scans;
- BOCPD for online inference;
- bounded-witness e-value or martingale scans, if theory is ready.

Use independent matched-null calibration, independent evaluation, block resampling for dependence, and report false-positive rate, detection power, delay/detection-within-window, localization error, and mechanism confusion. Do not tune thresholds on the evaluation trajectories.

10.7 Phase 5: long-horizon and pathwise extension

Only after one-step typed contrasts are reliable, test block or rollout laws $P(Y_{t+1:t+K} | H_t = h)$. Compare recursive and direct multi-horizon densities, conditional samplers, and paired pathwise contrasts. Vary horizon and coupling. The purpose is to learn whether finite SID effects compose or decay across time, not to introduce a separate latent mechanistic model.

10.8 Preregistered decision rules

1. No learned-backend experiment for a channel whose practical oracle adapter fails the zero baseline.
2. No downstream changepoint claim for a channel that fails static mechanism-confusion gates.
3. No winner based only on native loss; all winners must beat baselines on a common population estimand.
4. Independent generator, data, model-initialization, adapter, and evaluation seeds; exact seed intersections for pairwise comparisons.
5. A simple zero/no-effect estimator is always included for normalized derivative metrics.
6. One primary perturbation δ , corruption schedule, and metric per claim are fixed before evaluation.
7. Report uncertainty over generator seeds, failure counts, calibration, compute, and route agreement.
8. Preserve ridge, Gaussian/Student- t likelihood, MDN, SINDy, Granger/PCMCI/VAR-LiNGAM, MMD, energy, and appropriate changepoint baselines wherever their subproblem applies.

11 Research priorities and stop/go decisions

Direction	Priority	Why	Stop/go criterion
Finite typed SID	Highest	Most coherent generalization; real G1/G4 wins; model-agnostic	Continue if redesigned oracle G2 or G3 adapter beats zero, or if robust G1/G4 scaling advantage appears
Targeted Riesz/orthogonal channels	High	Directly addresses hard covariance/tail functionals and uncertainty	Continue only after oracle cross-fitted estimator is stable and beats direct moment baselines
Diffusion as backend	Conditional/high	Needed for sample-only/high-dimensional laws, not low-dimensional Gaussian tasks	Continue if it wins a home-field task or enables a unique validated estimand
SBTG-to-SID reconstruction	Medium	Theoretically connects existing ideas and may yield local structure	Continue if oracle $K_v \rightarrow \ell_v$ reconstruction is stable; otherwise retain as screen
Channel-specific changepoints	Medium after static gates	Strong mean/tail results; practical downstream use	Advance only validated channels with matched-null and confusion control
More SID-Q sweeps	Low	Current objective parity is decisive on tested parametric tasks	Resume only for a new intractably normalized model or theory-motivated estimator
Generic gating mixed derivative	Stop/redesign	Every learned readout loses to zero	Require controlled paired identification or explicit gated parameterization first

11.1 What would count as an impactful result?

At least one of the following should be demonstrated:

- a high-dimensional non-Gaussian conditional law where flexible-backend FINITE-SID beats feasible likelihood, direct moment regression, and generic two-sample methods on a common typed sensitivity;
- a sample-only setting where the framework reliably identifies a localized channel that density-based baselines cannot compute and direct empirical differences cannot estimate efficiently;
- a multi-contrast amortization result where one predictive generator plus SID adapters is substantially more accurate or efficient than separately trained targeted models;
- a calibrated changepoint system that both detects and correctly labels mean/dispersion/tail/interaction changes with controlled confusion under dependence;
- a theorem and experiment showing stable recovery of a centered SID witness from the SBTG mixed operator in a nontrivial class.

Without one of these, the framework may remain a useful organizational language but not yet a competitive method.

12 Recommended paper-level narrative

A future external paper should not claim that score matching generally outperforms likelihood or that a joint score reveals a causal mechanism. A defensible narrative is:

1. Define predictive-law sensitivity through exact finite signed contrasts and typed channel functionals.
2. Show that normalized densities, classifiers, scores, and samplers are interchangeable computational interfaces to the same estimand.
3. Establish boundedness, stability, and uncertainty results for the finite witness and its channel readouts.
4. Demonstrate oracle adapter validity, then learned-backend performance, in regimes where direct mean/variance models are insufficient.
5. Use SBTG as a related gradient representation and changepoints as a downstream inference application.
6. Be explicit that causal or mechanistic interpretations require extra assumptions or controlled perturbations.

The strongest conceptual contribution is not a new loss. It is a typed, support-aware interface between predictive generative models and scientifically interpretable distributional sensitivities, together with tests that determine whether the interface is valid.

13 Frozen July 13 empirical adjudication

Bottom line. Proper full-law fit is not sufficient for the hardest new SID target. Flexible autoregressive and flow densities learn G8’s static multimodal geometry extremely well by held-out likelihood, but retain only 3–19% of the held-out information about its history-controlled mode weights and their samplers recover at most about 10% of the corrected fourth-order effect. A classifier-guided midpoint generator raises EDM’s median effect slope from 0.010 to 0.398 but remains badly attenuated, and explicit endpoint experts reach only 0.336. These interventions separate missing global mass from missing fourth-order generator geometry. In contrast, a balanced finite classifier trained on the declared perturbation reaches Bayes AUC and 1–3% typed-effect error. Across the wider suite, direct ratios are the most reliable clean tangent route, while diffusion’s accurate noisy response score frequently fails to transfer to an accurate history tangent. The data therefore isolate an objective/estimand mismatch, not simply insufficient signal or an unstable sampler.

13.1 Frozen design and validation

The core run crossed eight exact generators, all applicable models, two generator seeds, two data seeds, and two model seeds. All 784 expected cells completed; all 784 case records were valid; there were zero case failures and zero

metric failures. The run used 8,000 training examples, a fixed 1,600-case validation set, 512 test cases, one CPU thread, and one job at a time. The eight crossed cells per generator–model pair measure data and initialization variability, but there are only two independent generated systems. Consequently all intervals in this subsection are developmental paired-bootstrap summaries, not confirmatory system-level inference.

The evaluation contract keeps five endpoints distinct:

1. fair energy score for conditional samples;
2. NLL only for models with an exact normalized density;
3. clean history-tangent NRMSE for direct normalized or ratio routes;
4. noisy response-score NRMSE at standardized corruption 0.12;
5. noisy history-tangent NRMSE for diffusion, separated by its centering route.

The zero-tangent estimator has NRMSE one. The frozen bounded-energy numerical rows are retained for audit but excluded from the primary fair-energy ranking because that run used dependent finite-pool SIR. The implementation has since been replaced by exact i.i.d. rejection sampling with proposal caps and acceptance diagnostics; this repairs future evaluation but does not retroactively change the frozen table.

13.2 Predictive-law fit: real flexibility gains, but metric-dependent

Table 3 gives the best eligible route under the two common law metrics. Energy values are mean paired percentage changes from Gaussian NLL; NLL values are paired nats per response vector. Negative is better.

Law	Best fair-energy route	Change	Best exact-density route	NLL change
G1 Gaussian sanity	Gaussian-anchored EDM	−0.41%	autoregressive MDN	−0.118
G2 covariance	autoregressive MDN	−0.66%	autoregressive MDN	−0.227
G3 matched skew	autoregressive MDN	−0.44%	autoregressive MDN	−0.665
G4 separated modes	autoregressive MDN	−2.25%	autoregressive MDN	−2.556
G5 near manifold	autoregressive Transformer	−4.98%	autoregressive Transformer	−24.336
G6 rough history	Gaussian NLL	0.00%	Gaussian NLL	0.000
G7 concentrated support	flow matching	−1.18%	autoregressive MDN	−1.228
G8 functional shape	affine flow	−1.89%	autoregressive Transformer	−18.343

Table 3: Best eligible law-fit routes in the frozen core panel.

Three conclusions are important. First, the flexible normalized models clearly learn non-Gaussian structure: relative to Gaussian, paired NLL improves in every cell by about 2.56 nats on G4, 24.34 on G5, and 18.34 on G8 for the winning models. Second, fair energy sees only small improvements on G4/G7/G8 and can therefore understate the value of fitting mode topology. Third, likelihood and sample geometry can disagree: on G5 the affine flow improves NLL by 23.24 nats but makes energy score 14.0% worse. Neither metric alone is a sufficient SID gate.

The stabilized EDM sampler is not a universal law winner. Relative to Gaussian its mean paired energy changes are −1.78% on G4, −0.93% on G7, and −1.42% on G8, but +6.03% on G5 and +1.37% on G1. Gaussian anchoring repairs much of the G5 loss (−3.25% versus Gaussian), but the autoregressive Transformer remains better (−4.98%). The legacy diffusion sampler is 43–130% worse than Gaussian across the suite despite often having the best noisy response-score estimate. Stabilization therefore repairs generation, not automatically derivative access.

The exact-rejection reevaluation closes the bounded-energy validity gap. All 64 frozen checkpoints complete with i.i.d. draws; median acceptance ranges from 0.136 to 0.431, and the largest realized proposal tensor contains 4,092 proposals under a 4,096 cap. Exact fair-energy medians for G1–G8 are respectively 1.507, 1.504, 1.775, 2.301, 1.209, 2.382, 2.778, and 0.715. These are competitive with Gaussian and flow on several laws, but still trail the strongest flexible frozen baseline on G4/G5/G7/G8. More decisively, the exact G8 quartic NRMSE is 1.000 with slope −0.00008. Exact sampling repairs the proper-score contract; it does not repair the contrast objective.

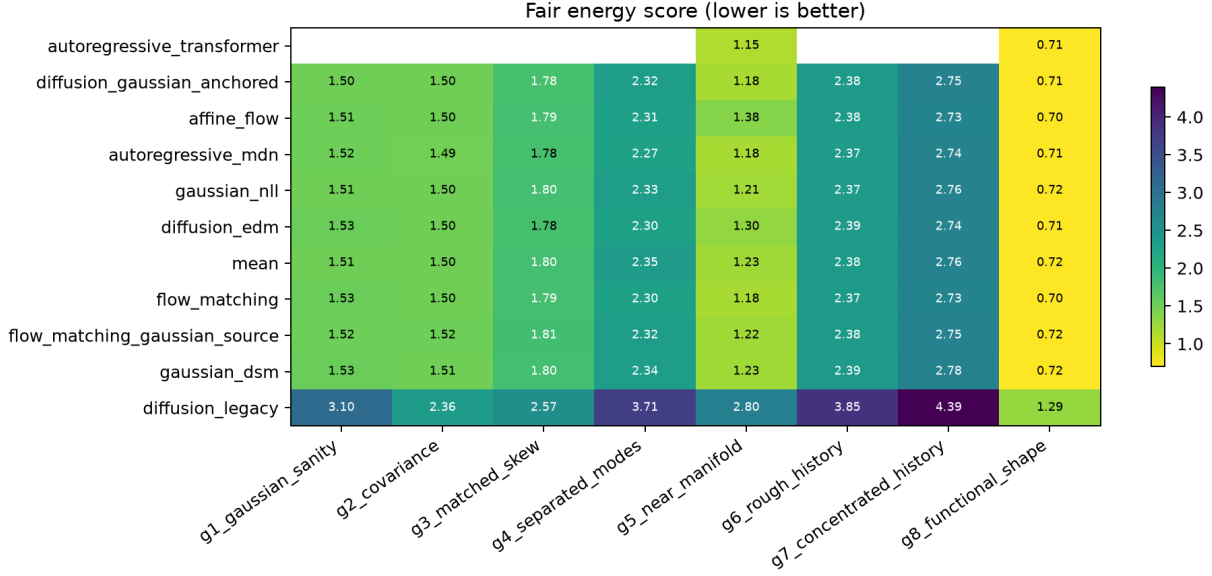


Figure 4: Frozen fair-energy medians. The bounded-energy row is excluded because that run’s SIR sampler did not provide i.i.d. draws. Absolute scores should be compared within a generator, not across response dimensions.

13.3 Clean SID: the ratio route wins most often, with sharp failures

Law	Best clean tangent route	Median NRMSE	Validity interpretation
G1 Gaussian sanity	ratio critic	0.532	valid in all eight cells
G2 covariance	ratio critic	0.582	valid in all eight cells
G3 matched skew	ratio critic	1.456	no route beats zero
G4 separated modes	ratio critic	0.424	only reliable route
G5 near manifold	ratio critic	0.585	Transformer second at 0.881
G6 rough history	ratio critic	0.658	all direct routes beat zero
G7 concentrated support	ratio critic	1.604	no route beats zero
G8 functional shape	autoregressive Transformer	0.529	ratio critic second at 0.616

Table 4: Clean history-tangent NRMSE; the zero estimator is one.

This is the strongest empirical support for the SID-driven framing. A direct ratio critic is the clean-tangent winner on six of eight laws and is particularly strong for separated mode occupancy (0.424) and the near-manifold mixture (0.585), where the Gaussian tangent is unusable (1.97 and 21.97). The normalized Transformer wins G8 (0.529). However, the ratio route is not a universal cure: G3 skew and G7 concentrated support remain invalid at 1.456 and 1.604. On the one-sided finite log-ratio metric, every existing G8 route is invalid: ratio critic 1.066, Transformer 1.071, affine flow 1.086, and Gaussian 1.134. Thus the law is learnable while the declared finite adaptation is not yet reliable.

13.4 Diffusion: response score and history SID empirically decouple

The legacy diffusion has the lowest noisy response-score NRMSE on every generator (0.216–0.753), yet after implementable model-sample centering its history-tangent NRMSE is invalid on G3, G4, G5, and G7. The corresponding medians are 2.390, 1.166, 1.125, and 5.852. EDM and Gaussian-anchored EDM generally have better samplers but worse mixed-history derivatives; on G8 their model-centered tangent NRMSEs are 1.246 and 0.850 versus 0.631 for legacy diffusion. This reverses their sample-quality ordering.

An independent five-system invalidity replication reaches the same location-versus-shape conclusion. Averaging

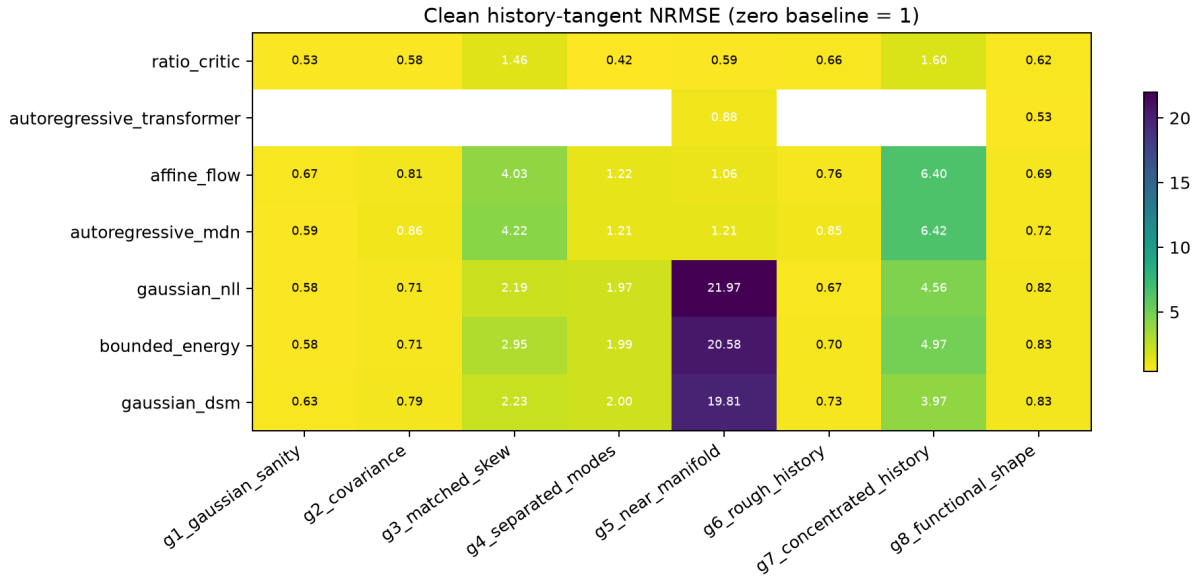


Figure 5: Clean history-tangent NRMSE with estimand and centering route held fixed. Values above one lose to the zero-tangent estimator.

three fixed Transformer coordinate orders, Transformer/diffusion tangent NRMSE is 0.497/0.751 on G1 location and 0.636/0.745 on G6 rough location, but 1.272/1.480 on G2 covariance, 1.014/3.852 on G3 skew, and 1.065/1.030 on G4 occupancy. The oracle clean-to-noisy gap is below 0.018, so small corruption does not explain the failure. A data-only history-Hyvärinen penalty also failed its frozen G2–G4 gate. These replications rule out the explanation that the core result is only one unlucky initialization or one EDM implementation choice.

In a separate 64-dimensional G5 development system, the converged Transformer has energy 1.1768, EDM 1.3226 (12.4% worse), and legacy diffusion 1.7072 (45.1% worse). Diffusion generates samples roughly 250 times faster, so it has a genuine computational advantage but no demonstrated quality advantage in that system.

13.5 Finite typed contrasts

The frozen finite panel uses the exact same source/checkpoint snapshot, three perturbation scales $\delta \in \{0.05, 0.12, 0.25\}$, and clean/noisy response laws. Its complete manifest contains 212,832 valid metric rows, 4,704 diagnostic rows, eight generators, 14 predictive-source labels, and seven finite routes; no metric row failed. The learned cells cross the same two generator, two data, and two model seeds as the core panel. One adapter seed is used, so these are exploratory medians rather than adapter-robust confidence intervals.

At the primary clean scale $\delta = 0.12$, the best learned typed-channel routes are:

Law	Best route/source	NRMSE	Interpretation
G1 location	coupled samples, anchored EDM	0.327	clear recovery
G2 covariance	coupled samples, flow matching	0.417	clear recovery
G3 matched skew	signed Riesz, legacy diffusion	1.003	no valid route at this scale
G4 separated modes	central log ratio, MDN	0.442	clear recovery
G5 near manifold	central log ratio, ratio critic	0.492	clear recovery
G6 rough history	coupled samples, flow matching	0.477	clear recovery
G7 concentrated support	coupled samples, flow matching	0.656	useful but variable
G8 functional shape	coupled samples, Gaussian-source flow	0.988	cubic dictionary is blind

This is more favorable to sampler backends than the infinitesimal-tangent panel. Direct common-random-number

sample moments let flow or stabilized diffusion lead G1, G2, G6, and G7; normalized or direct ratios lead G4 and G5. Full-witness reconstruction remains harder. The best witness NRMSEs are 0.508 (G1), 0.651 (G2), 0.342 (G4), 0.639 (G5), 0.724 (G6), and 0.698 (G7), while G3 and G8 remain approximately the zero witness. The signed Riesz dictionary is generally unstable and should not advance in its present form.

Perturbation scale is scientifically material. At $\delta = 0.25$, direct EDM samples recover the G3 typed skew channel at NRMSE 0.660 versus an oracle-sample Monte Carlo floor of 0.430; at $\delta = 0.12$ the same clean route is 1.042. At the noisy $\sigma = 0.12$ law and $\delta = 0.12$, EDM reaches 0.727. This is a real home-field hint for stabilized generative sampling, but it does not establish clean infinitesimal skew recovery: increasing δ changes the estimand, and response noising changes the law. More generally G4–G7 improve as the supported contrast grows, reinforcing the finite rather than infinitesimal formulation.

The process returned nonzero only after atomically writing the complete artifacts, when a post-write disk guard measured 3.87 GiB free against a 4.00 GiB floor. The manifest status, row counts, finiteness, and hashes were independently verified; the resource event is retained in the audit rather than relabeled as a scientific failure.

The G8 conclusion requires the analytically necessary fourth-order motif and exact frame-occupancy readouts. The current fixed response dictionary ends at cubic order and is a predeclared negative-control limitation rather than evidence that G8 has no signal.

The omitted target is not weak. For generator seeds 61 and 67, the analytic quartic constants are $D = 8.619$ and 7.583 . At control zero the exact central quartic effects are $5.170/4.548$ at $\delta = 0.05$, $5.162/4.542$ at $\delta = 0.12$, and $5.133/4.516$ at $\delta = 0.25$. A 25,000-draw-per-side oracle check at $\delta = 0.12$ recovers both effects within 0.78 Monte Carlo standard errors, while linear and quadratic motif contrasts stay near zero. Hence a learned method that misses the corrected quartic channel is missing a strong, exactly identified higher-order effect; a cubic dictionary simply never asked the right question.

A corrected frozen-checkpoint sample evaluation then asks that question directly. It covers 96 records with 12 histories and 512 draws per side for every eligible model; the oracle uses 2,048. The oracle quartic NRMSE is 0.117 and its median effect-calibration slope is 0.984. Every learned sampler is essentially the zero effect estimator:

Backend	Quartic NRMSE	Effect slope
autoregressive MDN	0.946	0.060
affine flow	0.949	0.051
EDM	0.994	0.006
autoregressive Transformer	0.994	0.007
ordinary flow matching	0.999	0.001
Gaussian NLL	1.000	0.000

Thus the large G8 likelihood gains come from learning the static multimodal path geometry, while the backends recover only 0–6% of its history-controlled fourth-order effect. This refines the diagnosis again: the original readout was blind, and the fitted predictive laws use too little of their capacity for the history-controlled mass change. A continuously varying history with one response per history could have contributed, so a separate factorial tests that hypothesis directly.

That 48-cell factorial crosses two independent systems, two model seeds, a balanced binary control, and a fixed-budget paired design with four outcomes per state/control cell. Binary randomization improves the strongest slope only to 9.5%; fixed-budget pairing does not rescue it:

Backend	Unpaired binary slope	Paired/replicated slope
autoregressive MDN	0.095	0.082
affine flow	0.046	0.050
autoregressive Transformer	0.049	0.024
EDM	0.013	0.016
flow matching	0.001	−0.001
Gaussian NLL	0.000	0.000

All 48 fits completed without failure. The paired lane holds total training rows at 8,000, so its additional within-state replication trades off against the number of unique states; it rules out that fixed-budget repair, not every possible replication regime. More importantly, the shape control is state-independent in G8, so persistent attenuation after a large supported -1 versus $+1$ contrast points toward objective/representation allocation rather than local support or a weak perturbation.

An exact noise-scale audit isolates why denoising is especially vulnerable on this law. Across four independent G8 systems and 5,000 exact draws per control side and system, the clean $+1$ versus -1 laws retain median Jensen–Shannon divergence 0.151 nats and Bayes AUC 0.770. Their response scores differ by only 3.86% of the score RMS, while the history-tangent RMS is 0.595. Noising initially exposes the inter-frame ambiguity region: relative response-score separation peaks at 13.56% near $\sigma = 0.20$. But at that same scale Jensen–Shannon divergence has already fallen to 0.0887 nats. By $\sigma = 0.60$, divergence is 0.00195 nats and history-tangent RMS is 0.0670. Thus the DSM signal has a narrow intermediate-noise window; ordinary averaging over noise levels is poorly aligned with preservation of the finite SID target.

G8 mode-weight signal across denoising scales

Four independent systems; medians from 5,000 exact draws per control side and system

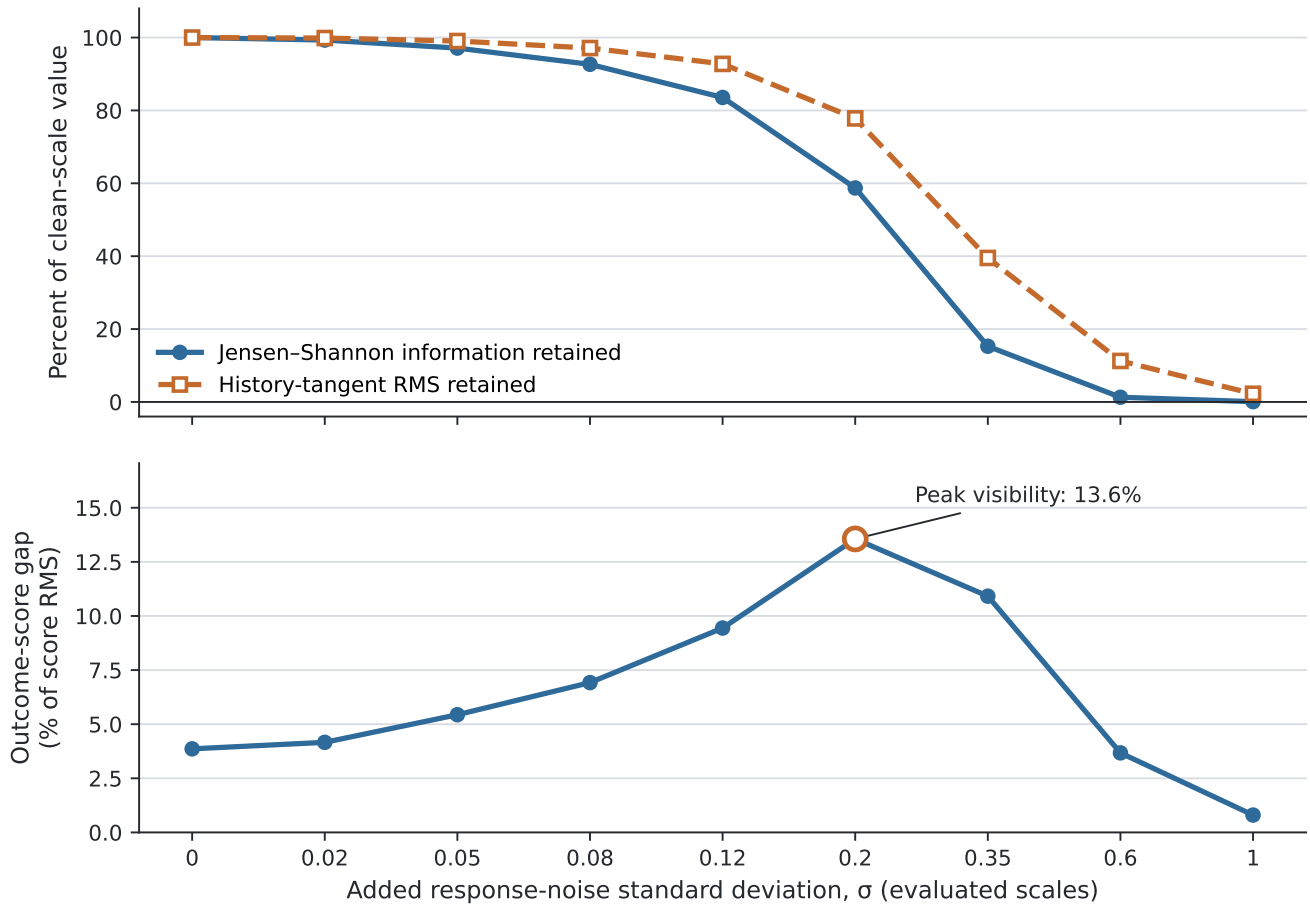


Figure 6: Oracle visibility of the G8 mode-weight contrast across denoising scales. Top: finite law information and history-tangent magnitude, normalized to the clean scale. Bottom: separation between the two conditional outcome scores relative to their RMS magnitude. Points are medians over four independent systems; each system uses 5,000 exact draws per control side.

The contrast itself is nevertheless learnable. A separate balanced-classification experiment trains directly on 4,000 responses per control side, tunes on 1,000, and evaluates on 10,000 held-out responses per side. It crosses four independent G8 systems and two data seeds per system. No classifier receives the control label as an input. With

equal class priors, its probability estimate is converted to the exact bounded finite witness $(2\hat{\eta} - 1)/\delta$ and then integrated against the quartic readout.

Estimator	AUC	Witness NRMSE	Median effect error
Bayes witness	0.768	0.000	0.33%
empirical typed difference	—	—	0.70%
MLP on motif coordinates	0.768	0.229	1.13%
typed quartic logistic	0.768	0.372	1.16%
fourth-order polynomial logistic	0.768	0.346	2.24%
MLP on raw path and baseline history	0.763	0.421	3.05%
quadratic motif logistic	0.508	1.002	99.64%
linear motif logistic	0.499	1.000	100.00%

This is the cleanest positive result in the new suite. The flexible contrast classifier reaches Bayes AUC and accurately recovers the typed effect; the raw 32-dimensional path model does nearly as well without the oracle motif basis. Linear and quadratic routes stay at chance and reproduce the exact negative controls. Full witness recovery remains harder than one projected functional—the motif MLP has witness NRMSE 0.229 despite only 1.13% typed-effect error—which is direct evidence for typed projection rather than unnecessary reconstruction of the entire law derivative.

The exact first/second-moment Gaussian shadow supplies the mandatory confusion control. Across the same four system seeds and two data seeds, the raw path/history classifier has median AUC 0.499 (range 0.496–0.505). Its absolute projected quartic effect is below 0.16% of the matched positive G8 effect in every cell. Thus neither a low-order mismatch nor accidental inclusion of the control label explains the positive G8 result.

This panel is developmental, not confirmatory: it has only four independent systems, uses clean histories, and studies the supported -1 versus $+1$ contrast. The motif and quartic variants use oracle centering/basis information; the raw path/history MLP is the relevant representation-agnostic comparator. Its 10,000-draw evaluation is larger than the sampler audit, although plug-in standard errors projected to 512 draws per side are still only 2.0% of truth for the motif MLP and 2.4% for the raw model. A final common-budget replication remains required.

A frozen-generator repair experiment then samples each fitted G8 control-zero midpoint law and reweights those draws with the same raw path/history classifier. It crosses two independent systems, two data seeds, two model seeds, and five generator families. The exact midpoint law with the learned classifier reaches median NRMSE 0.125 and slope 0.889; replacing the learned classifier by the Bayes posterior reaches NRMSE 0.013 and slope 1.001. Thus the reweighting identity and Monte Carlo procedure are numerically valid.

Frozen proposal law	Raw NRMSE	Raw slope	Guided NRMSE	Guided slope
EDM	0.991	0.010	0.602	0.398
flow matching	0.992	0.009	0.648	0.353
affine flow	0.945	0.055	0.661	0.340
autoregressive MDN	0.941	0.060	0.669	0.332
autoregressive Transformer	0.996	0.005	0.737	0.264

All 44 records, including four oracle-midpoint controls, completed successfully. Median guided effective sample fractions are 0.83–0.84 and classifier probabilities almost never saturate, so importance degeneracy is not the failure. Guidance recovers a real fraction of the missing mass effect for every proposal law, but none approaches the exact-midpoint control. In the error decomposition of the main text, the contrast head is useful while the Q -versus- M geometry term remains large. This rejects both extreme explanations: the generator is not irrelevant, and a classifier head alone is not a complete generative repair.

An 80-cell endpoint-expert factorial tests the complementary repair: enforce the global gate first, then fit a separate full-law expert to each supported endpoint. The total budget remains 8,000 rows (4,000 per expert), the declared control is removed from expert inputs, and the gate deterministically selects the $-1/+1$ expert. It crosses four systems, two data seeds, two model seeds, and five backends; every cell succeeds.

Endpoint expert	Fair energy	Quartic NRMSE	Quartic slope
EDM	0.704	0.665	0.336
affine flow	0.704	0.750	0.251
autoregressive MDN	0.714	0.823	0.179
flow matching	0.703	0.965	0.035
Gaussian NLL	0.718	1.000	0.000

Compared with shared balanced-control fits, explicit experts materially improve EDM (slope 0.013 to 0.336), affine flow (0.046 to 0.251), MDN (0.095 to 0.179), and flow matching (0.001 to 0.035). Yet they remain far below the direct finite classifier. This is the strongest architectural diagnosis in the suite: weak conditioning is one failure, while a full-law expert objective that underweights the fourth-order topology is another. A normalized gate is necessary here but not sufficient.

The frozen-anchor stochastic-interpolant factorial supplies a separate transport control. It reuses the exact core splits and Gaussian anchors, changes only $\beta_t = t$ versus t^2 , fixes diffusion amplitude one, and crosses the same two systems, two data seeds, and two model seeds on G1 and G8. All 32 fits succeeded.

Law	Schedule	Fair energy	Typed NRMSE	Typed slope
G1 Gaussian location	linear	1.513	0.338	0.908
G1 Gaussian location	squared	1.513	0.319	0.864
G8 fourth-order path	linear	0.717	0.993	0.0075
G8 fourth-order path	squared	0.710	0.999	0.0007

Typed columns use the G1 finite mean effect and the supported G8 quartic effect at $\delta = 1$. The 64/128/256-step G8 slopes agree to within about one percentage point. Squared β improves G8 fair energy but worsens the SID channel, while both schedules learn a nontrivial G1 mean effect. This is direct evidence against the two most convenient optimization explanations: the sampler is stable, and removing the initial response impulse does not expose the missing mass derivative.

Finally, an exact-density permutation audit removes sampling and quartic-readout noise. It measures the held-out log-density gain from supplying the correct shape control instead of a permuted control. The oracle gain is median 0.274 nats. The autoregressive MDN retains 19.4% of it, affine flow 12.4%, and the autoregressive Transformer only 2.9%; Gaussian NLL and Gaussian DSM retain zero. Thus the attenuation is already present in the fitted conditional density and is not created by the sampler or the fourth-order statistic.

14 Final answer to the strategic questions

Which of our methods works best? For estimating one scientifically declared law change, the best original method is now the direct finite classifier/ratio version of typed SID. A ratio critic is the clean-tangent winner on six of eight laws, and a balanced classifier recovers G8’s pure fourth-order path effect within 1–3% median error while linear and quadratic baselines recover none. For fitting an entire predictive law, the answer is problem-dependent: Gaussian NLL remains best on its home field, while MDNs and autoregressive Transformers make large NLL gains on multimodal and near-manifold laws. Those proper-law gains do not guarantee conditional sensitivity. Neither standalone diffusion/DSM nor flow matching is currently the best SID estimator.

Do the non-winners have hope? Yes, but only in roles consistent with what they identify. Diffusion, stochastic interpolants, and function-space score models have a credible role as whole-path, within-mode, or sample-only generators. They do not have a credible standalone route to history-controlled masses on separated components; that regime requires a normalized mixture gate, finite classifier, ratio, or bounded energy head. Stable Target Field, topology-window noise weighting, EDM preconditioning, and stochastic interpolants can reduce variance or transport difficulty, but cannot create the missing component constants. The completed interpolant factorial confirms that boundary: squared β improves G8 fair energy but its quartic slope is only 0.0007 at 128 steps, with the same conclusion at 64/256 steps. SBTG has hope on connected, well-conditioned laws as a localizer or as a

centered Poisson/Hodge reconstruction with a proved stability constant. It should not lead on separated mode weights. Covariance and skew channels remain open because their existing adapters fail even with oracle access.

Where are we going? Toward a contrast-first hierarchy: (i) direct finite classifier and projected-moment baselines for every declared readout; (ii) normalized mixture-of-experts or bounded ratio/energy heads for global mass; (iii) diffusion, flow, or function-space operators only for residual path geometry that simpler normalized models cannot fit; and (iv) joint history/future measurement stress tests only after the clean-law channel passes. Exact rejection sampling makes the bounded energy backend valid but does not make it the winner; post-hoc classifier guidance raises EDM’s G8 slope only to 0.40, and explicit endpoint experts reach 0.34. The decisive next experiment is a common-budget mixture-gated, contrast-aligned diffusion/flow versus direct classifier, MDN, autoregressive likelihood, kernel two-sample, and empirical typed difference on several independent high-dimensional path laws. Static validity and route agreement precede changepoints or real neural interpretation.

A Reproducibility and evidence map

This synthesis concerns theory and synthetic data only. Real neural-data performance is intentionally outside scope. The principal local evidence sources are:

- `SYNTHETIC_METHODS_REPORT_2026-07-12.md`: prior adjudication and exact result summaries;
- `analysis/synthetic_methods_adjudication_2026-07-12.ipynb`: executed analysis notebook;
- `analysis/synthetic_methods_adjudication/method_taxonomy.csv`: method inventory;
- `neuromod_benchmark/configs/objective_adjudication_20260712.yaml`: matched objective configuration;
- `neuromod_benchmark/outputs/objective_adjudication_20260712/`: 250-case matched objective outputs;
- `history_tangent_benchmark/results/revised_note_v1_finite_contrast_20260712_seed5004` through `seed5013`: ten-seed finite contrast outputs;
- `neuromod_benchmark/outputs/rethink_suite_20260712_v2/`: 24-neuron E1–E5 run with neural and score-model panel (840 fits);
- `neuromod_benchmark/outputs/rethink_suite_20260712_n32_fast/`: 32-neuron scale confirmation (672 fits);
- `neuromod_benchmark/outputs/rethink_suite_20260712_sid_stability/`: 432-fit SID regularization/corruption diagnostic;
- `neuromod_benchmark/docs/THEORY.md` and `neuromod_benchmark/docs/METRIC_CONTRACT.md`: estimands, claim ladder, and metric contract;
- `tmp/pdfs/SBTG_Revised_Technical_Note_July_2026.txt`: revised predictive-generator and finite-witness theory note.
- `history_tangent_benchmark/results/stable_sid_core_20260713/`: frozen 784-cell core run;
- `history_tangent_benchmark/results/stable_sid_finite_20260713/`: complete 212,832-row finite-contrast run and manifest;
- `analysis/g8_typed_classifier_benchmark_2026-07-13/`: eight-cell direct finite-classifier audit;
- `analysis/g8_gaussian_shadow_negative_control_2026-07-13/`: eight-cell matched-moment classifier negative control;
- `analysis/g8_classifier_guided_generator_2026-07-13/`: 44-record frozen-generator guidance experiment;
- `analysis/g8_endpoint_expert_decomposition_2026-07-13/`: 80-cell contrast-aligned endpoint-expert factorial;
- `analysis/focused_stochastic_interpolant_2026-07-13/`: 32-fit frozen-anchor schedule and sampler-resolution factorial;
- `analysis/exact_bounded_energy_reevaluation_2026-07-13/`: 64-checkpoint exact-rejection reevaluation;

- `analysis/g8_score_visibility_audit_2026-07-13/`, `analysis/g8_control_information_audit_2026-07-13/`, and `analysis/g8_control_design_factorial_2026-07-13/`: score visibility, exact control information, and control-design audits.

Software checks at the time of final synthesis passed 103 history-tangent tests, including analytic exact-rejection, G8 geometry/readout/shadow, and stochastic-interpolant protocol tests, with three pre-existing Transformer warnings. The prior neuromod benchmark suite passed 247 tests. Passing tests establishes implementation consistency, not scientific identification; the oracle and baseline gates in this report are stricter.

B Compact claim ladder

Claim type	What is supported	What is not implied
Predictive law	Held-out conditional distribution fit	Correct derivatives or physical mechanism
Finite contrast	Difference between declared supported histories	Infinitesimal effect outside the stencil; causal intervention
Typed channel	Change in a declared response feature	Other channels or a unique mechanism
Localization/graph	Reduced-form predictive dependence support	Causal edge under hidden confounding
Changepoint	Calibrated detection/localization of a statistic change	Mechanism label unless confusion gates pass

References

- [1] A. Hyvärinen. Estimation of non-normalized statistical models by score matching. *Journal of Machine Learning Research*, 6:695–709, 2005. <https://jmlr.org/papers/v6/hyvarinen05a.html>
- [2] P. Vincent. A connection between score matching and denoising autoencoders. *Neural Computation*, 23(7):1661–1674, 2011. https://doi.org/10.1162/NECO_a_00142
- [3] Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole. Score-based generative modeling through stochastic differential equations. *International Conference on Learning Representations*, 2021. <https://arxiv.org/abs/2011.13456>
- [4] T. Kanamori, S. Hido, and M. Sugiyama. A least-squares approach to direct importance estimation. *Journal of Machine Learning Research*, 10:1391–1445, 2009. <https://jmlr.org/papers/v10/kanamori09a.html>
- [5] A. Gretton, K. M. Borgwardt, M. J. Rasch, B. Schölkopf, and A. Smola. A kernel two-sample test. *Journal of Machine Learning Research*, 13:723–773, 2012. <https://jmlr.org/papers/v13/gretton12a.html>
- [6] S. L. Brunton, J. L. Proctor, and J. N. Kutz. Discovering governing equations from data by sparse identification of nonlinear dynamical systems. *Proceedings of the National Academy of Sciences*, 113(15):3932–3937, 2016. <https://doi.org/10.1073/pnas.1517384113>
- [7] J. Runge, P. Nowack, M. Kretschmer, S. Flaxman, and D. Sejdinovic. Detecting and quantifying causal associations in large nonlinear time series datasets. *Science Advances*, 5(11):eaau4996, 2019. <https://doi.org/10.1126/sciadv.aau4996>
- [8] A. Hyvärinen, K. Zhang, S. Shimizu, and P. O. Hoyer. Estimation of a structural vector autoregression model using non-Gaussianity. *Journal of Machine Learning Research*, 11:1709–1731, 2010. <https://jmlr.org/papers/v11/hyvarinen10a.html>
- [9] R. Pamfil, N. Sriwattanaworachai, S. Desai, P. Pilgerstorfer, K. Georgatzis, P. Beaumont, and B. Aragam. DYNOTEARS: Structure learning from time-series data. *Proceedings of AISTATS*, PMLR 108:1595–1605, 2020. <https://proceedings.mlr.press/v108/pamfil20a.html>
- [10] V. Chernozhukov, W. K. Newey, and R. Singh. Automatic debiased machine learning of causal and structural effects. *Econometrica*, 90(3):967–1027, 2022. <https://doi.org/10.3982/ECTA18515>

- [11] R. Killick, P. Fearnhead, and I. A. Eckley. Optimal detection of changepoints with a linear computational cost. *Journal of the American Statistical Association*, 107(500):1590–1598, 2012. <https://doi.org/10.1080/01621459.2012.737745>
- [12] R. P. Adams and D. J. C. MacKay. Bayesian online changepoint detection. Technical report, 2007. <https://arxiv.org/abs/0710.3742>
- [13] A. K. Menon and C. S. Ong. Linking losses for density ratio and class-probability estimation. *Proceedings of ICML*, PMLR 48:304–313, 2016. <https://proceedings.mlr.press/v48/menon16.html>
- [14] T. Karras, M. Aittala, T. Aila, and S. Laine. Elucidating the design space of diffusion-based generative models. *Advances in Neural Information Processing Systems*, 35, 2022. <https://arxiv.org/abs/2206.00364>
- [15] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le. Flow matching for generative modeling. *International Conference on Learning Representations*, 2023. <https://arxiv.org/abs/2210.02747>
- [16] J. H. Lim, N. B. Kovachki, R. Baptista, et al. Score-based diffusion models in function space. arXiv:2302.07400, 2023. <https://arxiv.org/abs/2302.07400>
- [17] Y. Xu, S. Tong, and T. Jaakkola. Stable Target Field for reduced variance score estimation in diffusion models. *International Conference on Learning Representations*, 2023. <https://arxiv.org/abs/2302.00670>
- [18] M. Kolloviah, M. Lienen, D. Lüdke, L. Schwinn, and S. Günnemann. Flow matching with Gaussian process priors for probabilistic time series forecasting. arXiv:2410.03024, 2024. <https://arxiv.org/abs/2410.03024>
- [19] Y. Chen, M. Goldstein, M. Hua, M. S. Albergo, N. M. Boffi, and E. Vanden-Eijnden. Probabilistic forecasting with stochastic interpolants and Föllmer processes. *Proceedings of ICML*, PMLR 235:6728–6756, 2024. <https://proceedings.mlr.press/v235/chen24n.html>